

2

Current, Voltage and Resistance

In this chapter, we will learn about

- electric current, voltage and resistance
- Ohm's law
- the equivalent resistance of resistors connected in series and in parallel
- electrical power

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2.1 Electric current

1 Definition of electric current

DSE goal

Define electric current as rate of flow of electric charges.

Flipped classroom

Watch a video and answer the questions.

Video

→ This video demonstrates a neon bulb being lighted up by static charges.

You can light up a neon bulb by static charges (Fig 2.1a).



① Rub a plastic folder with a cloth to charge it.



② Wear a glove and put a metal ruler on the folder.



③ Hold one end of the bulb with a bare hand and let the other end touch the ruler.

Fig 2.1a *Lighting up a neon bulb.*

When the metal ruler is put on the charged folder, charges are induced on the ruler. When the bulb touches the ruler, charges flow through the bulb and the bulb gives a flash. We say that an **electric current** passes through the bulb. The size of the electric current depends on how fast the electric charges flow.

An electric current is the rate of flow of electric charges.

Mathematically, current I is defined as the total amount of charges flowing through a cross-section of a conducting wire per unit time (Fig 2.1b).

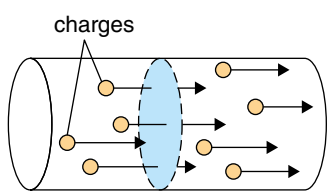


Fig 2.1b *Charges flowing through a wire.*

The unit of current is the **ampere** (**A**). A current of 1 A means that 1 C of charge passes through the cross-section every second.

$$I = \frac{Q}{t}$$

current (A) — I; charge (C) — Q; time (s) — t

1 A = 1 C s^{-1} ▶

Q is always taken as positive.

electric current 電流　　ampere (A) 安培

Some typical current sizes are given below (Fig 2.1c). In general, electrical appliances for heating or cooling draw a relatively large current.

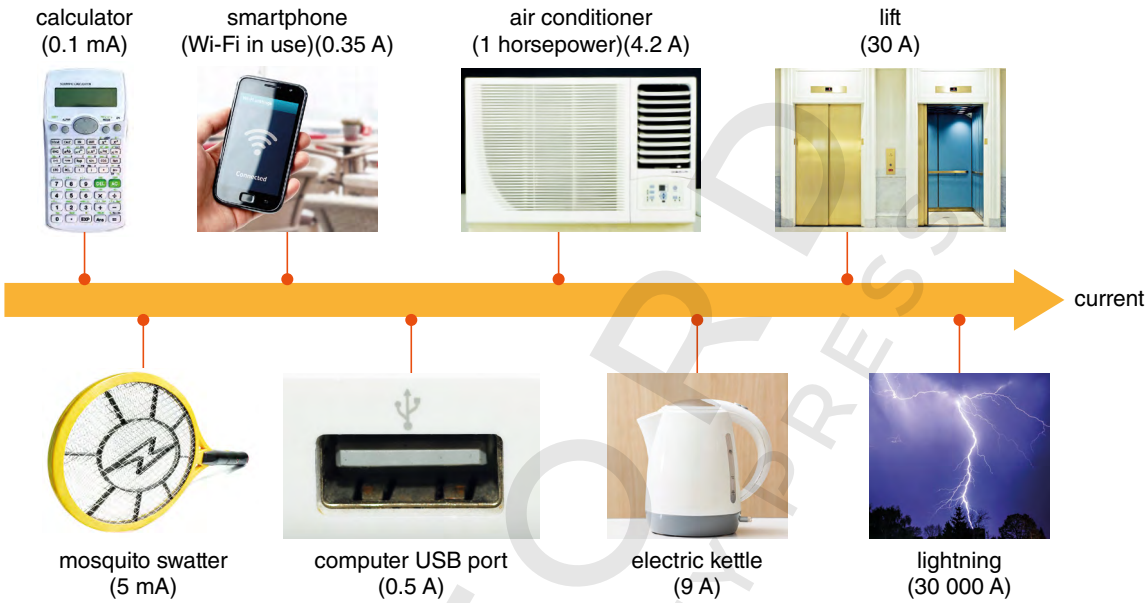


Fig 2.1c *Some typical current sizes.*

Example 1 | Current through an electric car

An electric car is charged by a current of 80 A. Find the amount of charge Q flowing through the car in 2 minutes.



Solution

$Q = It = 80 \times (2 \times 60) = 9600$ C

The amount of static charge in daily life is about 10^{-9} C. ▶

Know more

André-Marie Ampère (1775–1836)

André-Marie Ampère was a French physicist. He discovered the interaction between two wires carrying currents, which became the foundation of *electromagnetism*. The unit of current is named after him.



OUP eResources

André-Marie Ampère 安培　　electromagnetism 電磁學

DSE goal

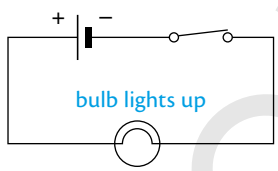
State the convention for the direction of current.

2 Circuit and current direction

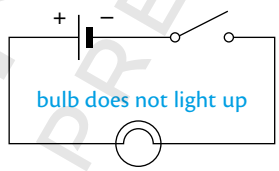
Refer to Figure 2.1a (p.56). If we want to keep the bulb lighting up, we will need to connect it to an electric source (e.g. a battery) via a closed circuit. Let us have a short revision on the circuit.

Pre-study **Basics of the circuit**

- A power supply only provides energy to drive the charges in the circuit. It does not provide charges.
- A closed circuit provides a closed path for electric currents to flow through. Conversely, no current flows in an open circuit.



A closed circuit.



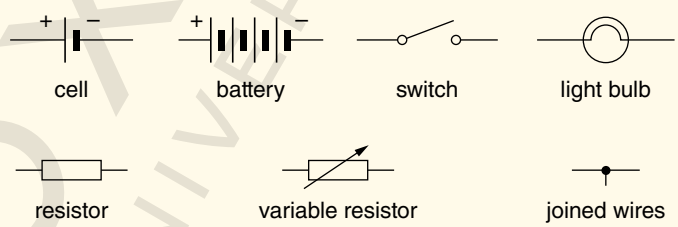
An open circuit.

For Ss studying chemistry, Ts may remind them that they have learned about the difference in electric potential of materials at the two electrodes of a chemical cell. The difference is due to the electrodes having different tendencies to lose or gain electrons.

Skill *Drawing*

Drawing circuit diagrams

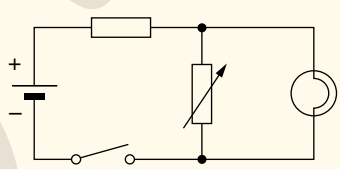
- Use circuit symbols to represent circuit components.



A battery can be a single cell, or it can consist of several cells connected together. ▶

→ Teaching notes and exercises are available on OUP web.

- Draw each component in the same position as it appears in the circuit.
- Draw the connecting wires as straight lines.



circuit symbols can be drawn vertically or horizontally

In a circuit, the flowing charges can be positive (+) or negative (–), depending on the materials of the circuit. Positive charges flow towards the negative terminal of a battery. In metals (e.g. connecting wires), negatively charged electrons flow in the opposite direction.

For practical purposes, the direction of current is defined as the direction in which positive charges flow, i.e. from the positive terminal to the negative terminal of the battery via the circuit. This is called the **conventional current direction** (Fig 2.1d on p.59).

Know more

Flow of charges

Scientists first thought that current is a flow of positive charges from the + terminal to the – terminal of a battery via a circuit. Afterwards, it was found that the current in metals is in fact a flow of electrons in the opposite direction. Moreover, in a conducting solution like tap water, positive ions and negative ions flow in opposite directions to form a current.

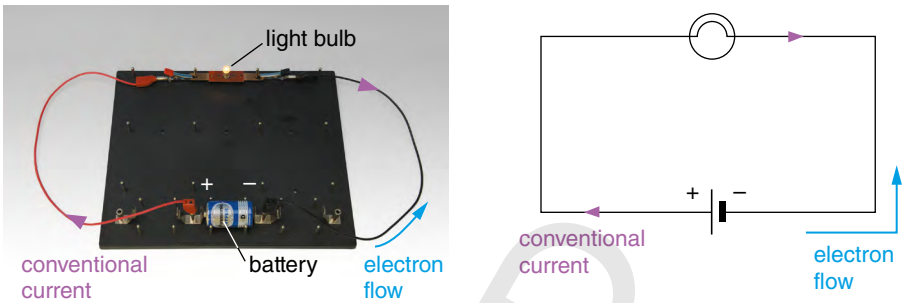


Fig 2.1d *The conventional current and the electron flow are in opposite directions.*

By convention, current flows from the positive terminal to the negative terminal of a battery via a circuit.

Right and wrong

✗ A circuit becomes charged when current flows.

✓ Current is a flow of charges (e.g. free electrons) already present in the circuit. The net charge of the circuit remains zero whether current flows or not.

Quick check 1

E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 Current flows in a circuit. Complete the following table.

	I	*Q*	*t*
(a)	A	1 C	2 s
(b)	4 A	C	1 h
(c)	500 mA	3000 C	s

2 *True or false:*

(a) A battery supplies charges to the circuit. (T/F)

(b) Current can flow continuously in an open circuit. (T/F)

(c) The current direction is opposite to the direction of electron flow. (T/F)

(d) A current of 1 A means that 1 electron passes through the cross-section of the conducting wire every second. (T/F)

3 1 C =

A 1 A m B 1 A m^{-1}

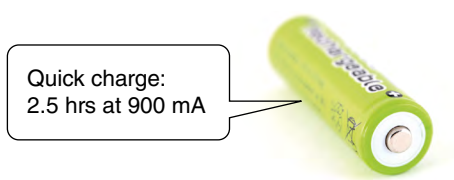
C 1 A s D 1 A s^{-1}

Learn by practice 1

1 A current of 20 mA passes through a resistor. Find the time needed for 1 C of charge to flow through the resistor.

2 In a circuit, 1.25×10^{19} electrons flow through a light bulb in 4 s. Find the current through the bulb. The charge of an electron is 1.60×10^{-19} C.

3 A rechargeable cell is fully charged at 900 mA in 2.5 hours. Estimate the total amount of charge that flows through the cell in this process.



DSE goal
Measure I in simple circuit.

3 Measuring current

We usually use an **ammeter** (Fig 2.1e) to measure current.

1 An ammeter should be connected **in series** with the component to be measured. It should **not** be connected **in parallel** with the component. Moreover, **never** connect an ammeter directly across a battery (Fig 2.1f).

This will damage the ammeter. ▶

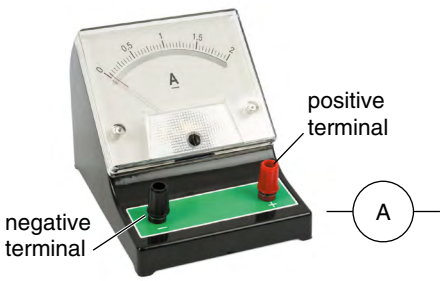


Fig 2.1e *An ammeter and its circuit symbol.*

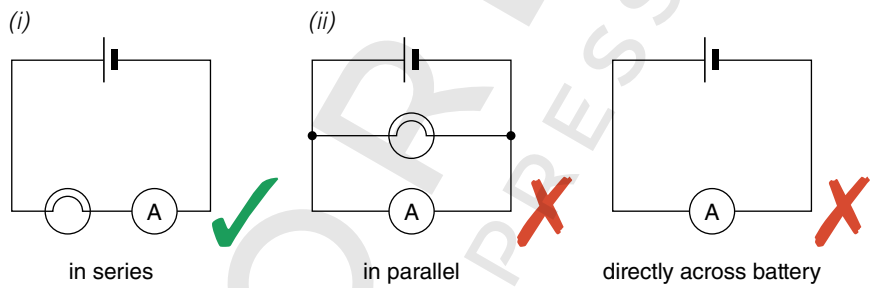


Fig 2.1f *(i) Correct and (ii) incorrect ways of connecting an ammeter.*

✎ Remind Ss that if an ammeter is connected in the wrong direction, its pointer may deflect to the other side and may be damaged.

2 The current should enter an ammeter via the red (positive) terminal and exit via the black (negative) one (Fig 2.1g).

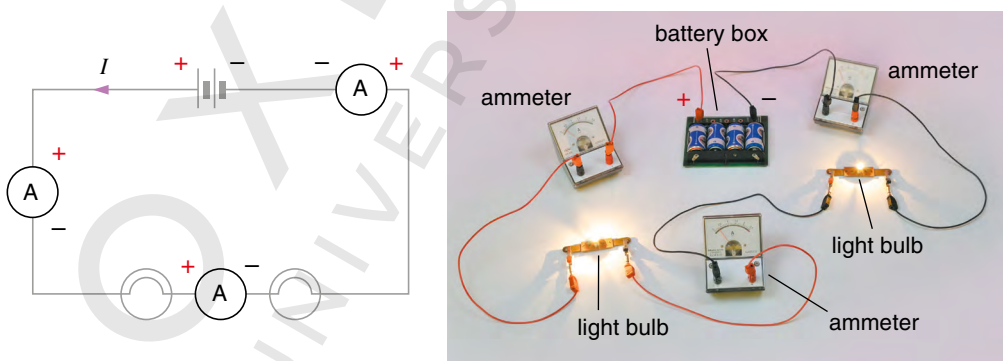


Fig 2.1g *Ammeters should be connected in the correct direction.*

A current sensor can record how the current changes with time. Its circuit symbol is the same as an ammeter. ▶

A *galvanometer* (Fig 2.1h) or a *current sensor* (Fig 2.1i) can be used to detect both the direction and size of a current. Like an ammeter, they should be connected in series with the component to be measured.

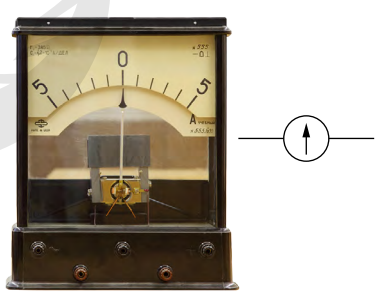


Fig 2.1h *A galvanometer and its circuit symbol.*



Fig 2.1i *A current sensor.*

ammeter 安培計　in series 串聯　in parallel 並聯　galvanometer 檢流計
current sensor 電流感應器

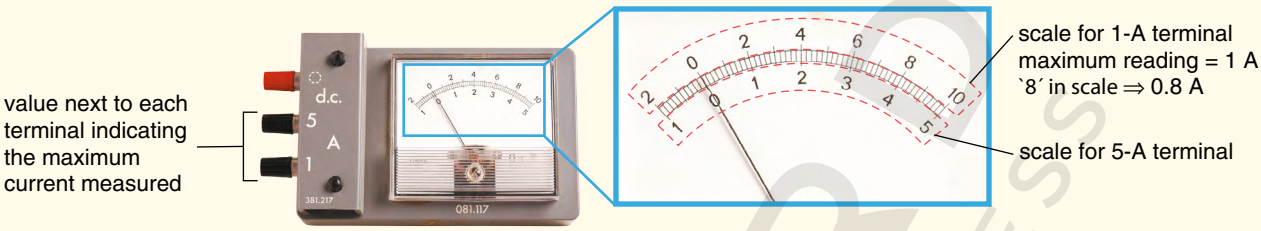
Simulation → The simulation provides practices for Ss to read a multi-range ammeter. It is available on OUP web in PC version (.exe) only.

Skill *Experiment*

Tips about using a multi-range ammeter

→ Teaching notes and exercises are available on OUP web.

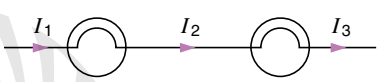
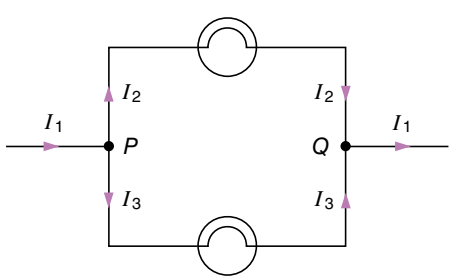
- A multi-range ammeter has more than one red/black terminal for different ranges of measurements. Look at the scale corresponding to the terminals in use to get the correct reading.



- Start measuring with the largest range first. This avoids the pointer from deflecting out of the scale, or the ammeter will be damaged.
- Switch to a smaller range for a more accurate reading if the current is within that range.

4 Current and conservation of charges

Since the total amount of charge flowing in a closed circuit is constant and charges will not accumulate at any point, the total charge flowing towards a point is always equal to the total charge leaving that point. This gives the following results.

Series connection	Parallel connection
 Current through any point must be the same: $I_1 = I_2 = I_3$	 Current entering a junction (P and Q) must be the same as that leaving the junction: $I_1 = I_2 + I_3$

1 The current is the same at all points in a series connection.
2 The current flowing into a junction is equal to the total currents flowing out from the same junction.

Example 2 Finding currents

Light bulbs are connected to a battery in different ways as shown.

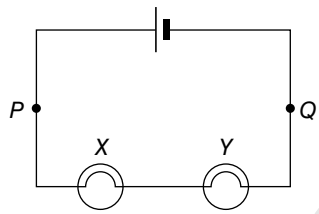


Fig a

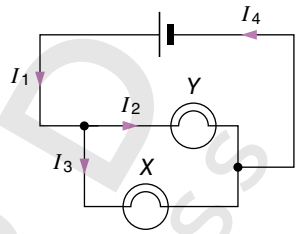


Fig b

(a) In Figure a, the current through point P is 1.5 A. Find the current through bulb X, bulb Y, point Q and the battery.

(b) In Figure b, if I_1 = 2.5 A and I_3 = 1 A, what are I_2 and I_4?

Solution

(a) Since they are all in series, the current through them is 1.5 A.

(b) $I_2 = I_1 - I_3 = 2.5 - 1 = 1.5$ A

$I_4 = I_1 = 2.5$ A

Alternatively,
$I_4 = I_2 + I_3$
$= 1.5 + 1$
$= 2.5$ A ▶

Google Forms Microsoft Forms

Quick check 2

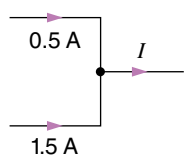
E-version in OUP Exercise Platform is also available.

1 Find the current I in the following cases.

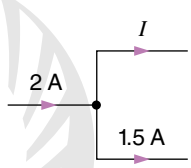
(a)



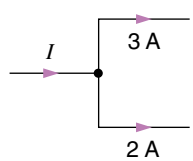
(b)



(c)

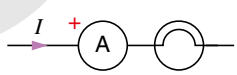


(d)

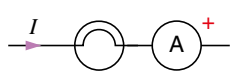


2 Which of the following ammeters could measure the current passing through the bulb? Put a tick in the corresponding box(es).

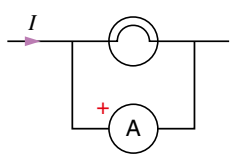
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Exercise 2.1

★**1** Metal wires *X* and *Y* have different diameters. They are connected in series to a battery (Fig a). Which of the following statements are correct?

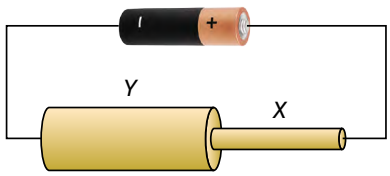


Fig a

(1) The current in *X* is equal to the current in *Y*.

(2) Current flows from *X* to *Y*.

(3) Both wires are negatively charged.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

2 Read the currents shown in the ammeters below.

(a)



(b)



3 The figure shows a simple circuit (Fig b). Redraw the circuit using circuit symbols. Indicate the + terminals of the battery and the ammeter.

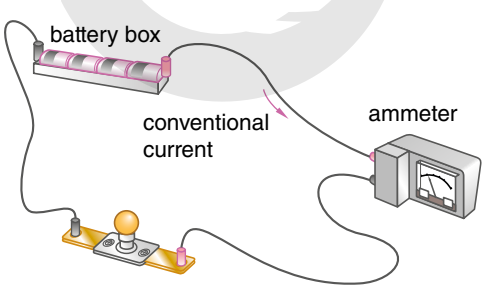


Fig b

4 Some light bulbs are connected to a battery (Fig c). A current of 3 A passes point *A* and a current of 2 A passes point *D*.

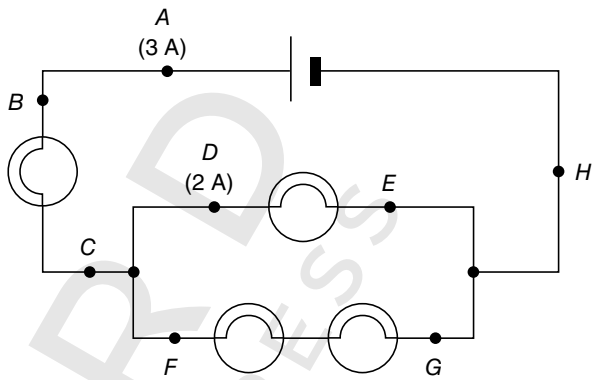


Fig c

(a) Draw arrows to the show the current direction at points *C*, *D* and *F*.

(b) Determine the size of current passing points *B*, *C*, *E*, *F*, *G* and *H*.

(c) Find the amount of charges passing point *A* in 5 minutes.

★**5** A circuit is shown in Figure d.

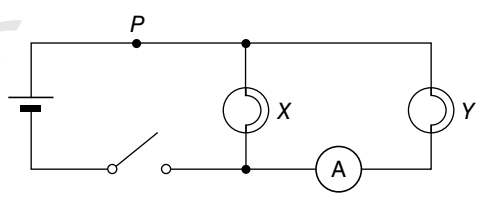


Fig d

(a) Are the two bulbs connected in series or in parallel?

(b) When the switch is closed, the current at point *P* is 2 A and the reading of the ammeter is 1.5 A. Find the current through bulbs *X* and *Y*.

(c) Draw lines in Figure e to connect the components according to Figure d.

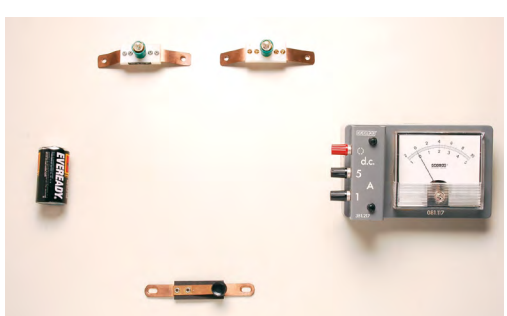


Fig e

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2.2 Voltage

1 Energy transformation in a circuit

DSE goal
Describe energy transformation in electric circuits.

Simulation

→ The simulation 'Model of a simple circuit' demonstrates how energy is transferred from the source to the load in a simple circuit by means of electrons flowing round the circuit.

→ The simulation 'Energy change in a circuit' shows the energy changes in a simple circuit.

The simulations are available on OUP web in PC version (.exe) only.

You may have seen the toy below (Fig 2.2a).



Fig 2.2a *Penguin slide.*

It can be used as an analogy of a circuit. The escalator, penguins and track represent the battery, charges and *load* (e.g. lamps) respectively (Fig 2.2b).

Remind Ss that we assume the current as a flow of positive charges.

Energy is conserved. When a battery 'provides' energy or when a load 'consumes' energy, energy is converted from one form to another.

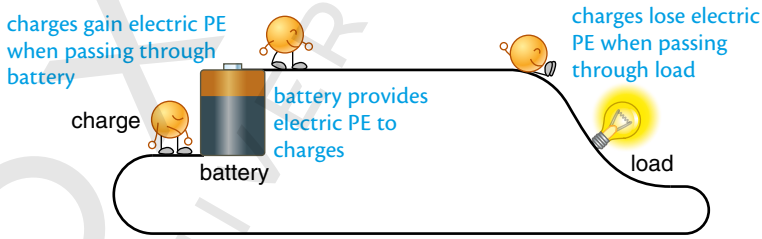
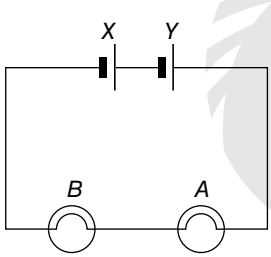


Fig 2.2b *How energy of charges changes in a circuit.*

The electric PE of the charges in a circuit can also be referred as electrical energy.

A circuit is an energy transfer system. Charges gain electric potential energy (electric PE) from the battery and then transfer the energy to the loads (Fig 2.2c).



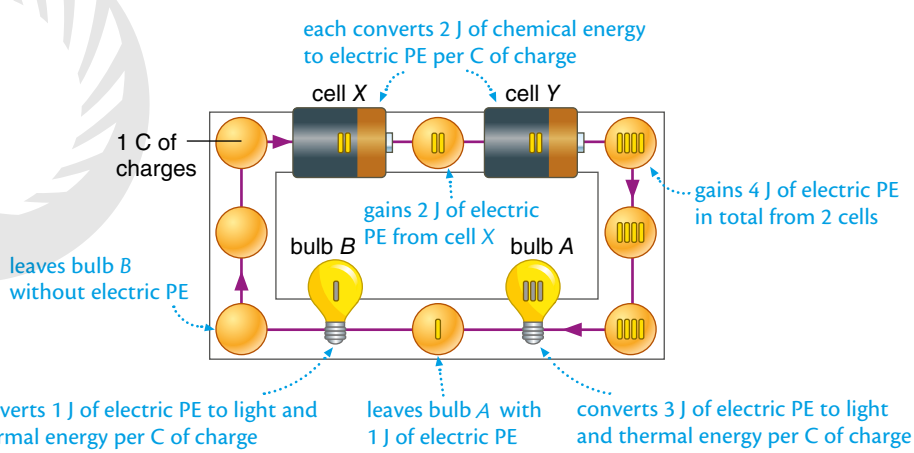


Fig 2.2c *Energy transformation in a circuit.*

Stress that bulb *A* and bulb *B* are of different ratings (or simply different brightness).

load 負荷

DSE goal

Define e.m.f.

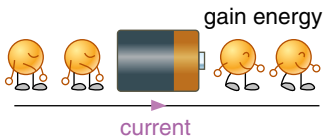
Define p.d.

DSE exam

18(1A)Q26, 20(1A)Q25

2 E.m.f., p.d. and voltage

In a circuit, an electric source (e.g. a battery) provides energy while a load consumes energy. We can use the quantities below to compare the amount of energy converted in these circuit elements.



a E.m.f. of a source

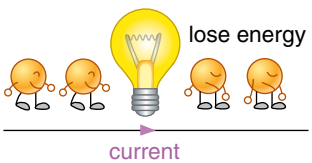
A source provides energy to charges so that they can flow steadily in a circuit. The **energy imparted by a source per unit charge** passing through it is called the **electromotive force** (**e.m.f.**) of the source. It is denoted by ε. Its unit is the **volt** (**V**).

Caution

Despite its name, electromotive force is not a force.

$1\ V = 1\ J\ C^{-1}$

electromotive force of source (V): $\varepsilon = \frac{E}{Q}$ — E: energy imparted by source (J); Q: charge (C)



b P.d. across a load

When charges pass through a load, some of their electric PE is converted to other forms of energy by the load. The **electric PE converted** to other forms **per unit charge** passing **between two points in the external circuit** is called the **potential difference** (**p.d.**) between the points. It is measured in V.

The part of a circuit outside the source is called the external circuit.

potential difference between two points (V): $\text{p.d.} = \frac{E}{Q}$ — E: electric PE converted by load (J); Q: charge (C)

The potential difference across a load is sometimes called the potential drop at the load.

It is important to make a clear distinction between the e.m.f. of a battery and the p.d. across a load.

In Figure 2.2c (p.64), the e.m.f. of the battery is 4 V. The p.d. across bulbs *A* and *B* are 3 V and 1 V respectively (Fig 2.2d).

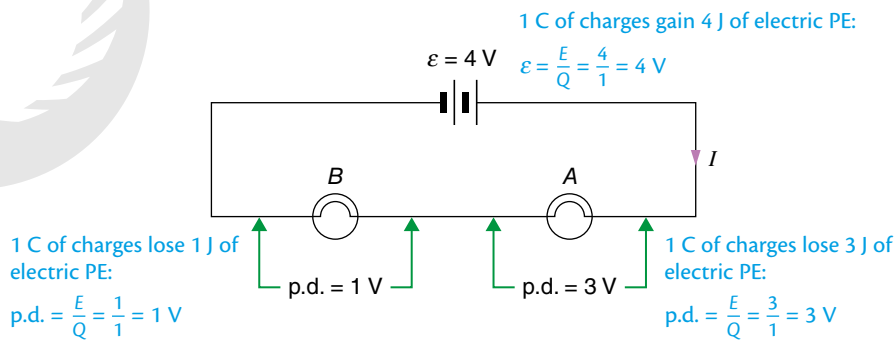


Fig 2.2d *E.m.f. of a battery and p.d. across light bulbs.*

electromotive force (e.m.f.) 電動勢 volt (V) 伏特 potential difference (p.d.) 電勢差

Know more

Electric potential

We usually choose the – terminal of the battery or any earthed points as the reference point of zero electric potential.

▸ p.109

Figure 2.2e illustrates the variation in the **electric PE per unit charge**, which is called the **electric potential**, along the circuit. The unit of electric potential is V. Note that:

1 The energy imparted by the battery is equal to the total energy consumed by all the loads.

2 The p.d. across two points is the **difference in electric potential** of the points. For example, the electric potential of point *X* is 0 V and that of point *Y* is 1 V, so the p.d. across them is 1 V.

The p.d. *V* across *X* and *Y* is also equal to the **difference** between the e.m.f. of the battery and the p.d. across *A*:
V = 4 – 3 = 1 V

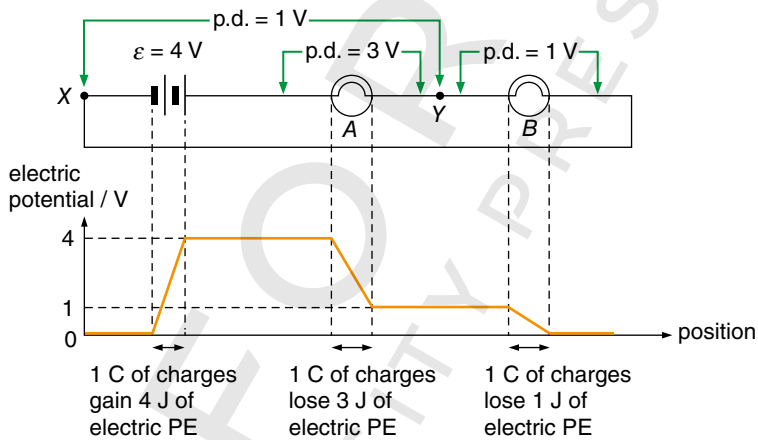


Fig 2.2e *The variation in electric potential along a circuit.*

Y has a higher potential than *X*, so current flows from *Y* to *X* in the external circuit.

3 A current is produced by a p.d. It flows **from a position of higher potential to a position of lower potential** in the external circuit.

C Voltage

We can use **voltage** *V* to mean **any change in electric PE per unit charge** in a circuit. Therefore, the voltage of a source is its e.m.f. and the voltage across a load is the p.d. across it.

voltage (V) $V = \frac{E}{Q}$ change in **electric PE** (J); charge (C)

Table 2.2a compares e.m.f., p.d. and voltage.

	E.m.f.	P.d.	Voltage
Applies to	source	load	source and load
Unit	V	V	V
Energy transformation	other forms → electric PE	electric PE → other forms	other forms ↔ electric PE

Table 2.2a *Comparison of e.m.f., p.d. and voltage.*

Know more

Alessandro Volta (1745–1827)



Alessandro Volta was an Italian physicist. He was a pioneer of electricity and invented the first battery. The unit volt is named in his honour.

OUP eResources

electric potential 電勢　voltage 電壓　Alessandro Volta 伏打

✎ Ask Ss to identify which are e.m.f. and which are p.d.

Figure 2.2f shows some typical values of voltage.

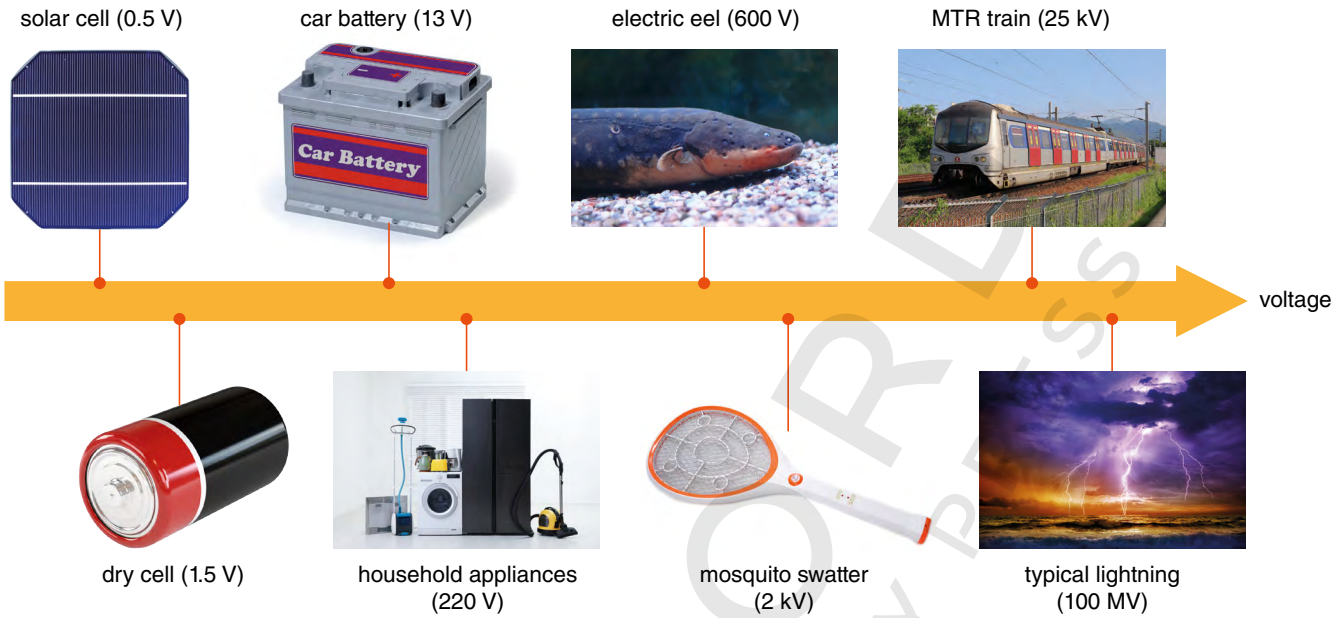


Fig 2.2f *Some typical values of voltage.*

✎ A current larger than 0.1 A can be fatal.

A high voltage is dangerous because it can generate a large enough current to cause injury or even death.



Fig 2.2g *Warning sign of high voltage.*

Example 3 Energy in a rechargeable battery

The capacity of a cell is the amount of charge supposed to gain energy from the cell. It is measure in ampere hour (A h).

A 1.2-V rechargeable battery has a capacity of 2950 mA h. This means that it can provide a current of 2950 mA for 1 hour.



(a) How many coulombs of charges can be driven by the fully-charged battery to flow in a circuit?

(b) How much energy is stored in the fully-charged battery?

Solution

(a) $Q = It = (2950 \times 10^{-3}\ \text{A})(3600\ \text{s}) = 1.06 \times 10^{4}\ \text{C}$

(b) $E = VQ = (1.2)(1.06 \times 10^{4}) = 1.27 \times 10^{4}\ \text{J}$

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DSE goal
Measure *V* in simple circuit.

3 Measuring voltage

A **voltmeter** (Fig 2.2h) is used to measure the voltage across two points. Note that:

1 It should be connected **in parallel** with the circuit component to be measured (Fig 2.2i).

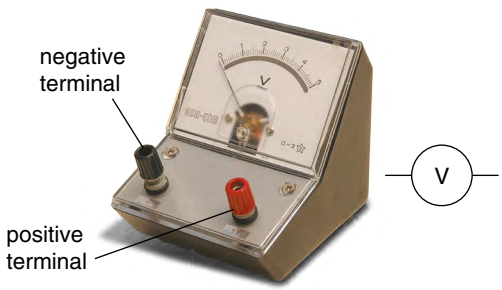


Fig 2.2h *Voltmeter and its circuit symbol.*

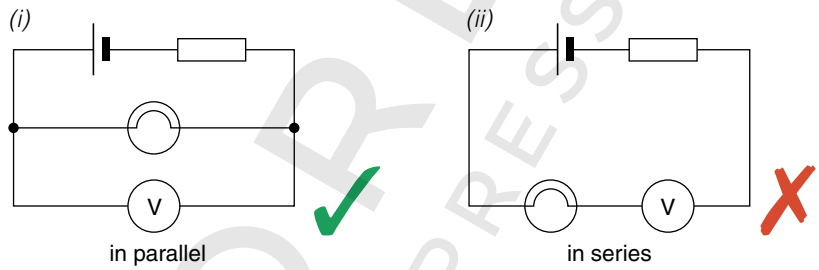


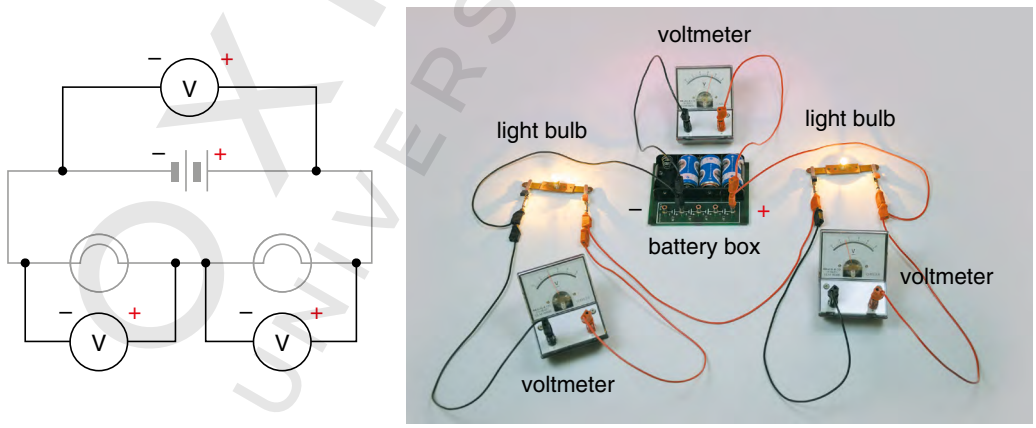
Fig 2.2i *(i) Correct and (ii) incorrect ways of connecting a voltmeter.*

Similar to ammeters, some voltmeters have more than one red/black terminal for different ranges of measurement. The Skill on p.61 can also be applied to these voltmeters.

Simulation

→ The simulation provides practices for Ss to read a multi-range voltmeter. It is available on OUP web in PC version (.exe) only.

▶ 2 Its red (positive) terminal should be connected to the positive terminal of the battery (Fig 2.2j).



▶ **Fig 2.2j** *Voltmeters should be connected in parallel in the correct direction.*

This simulation allows you to connect circuits freely and measure the voltage and current at different positions.

Simulation

A voltage sensor (Fig 2.2k) and a CRO (Fig 2.2l) can be used to record the variation of voltage over time.



Fig 2.2k *Voltage sensor.*

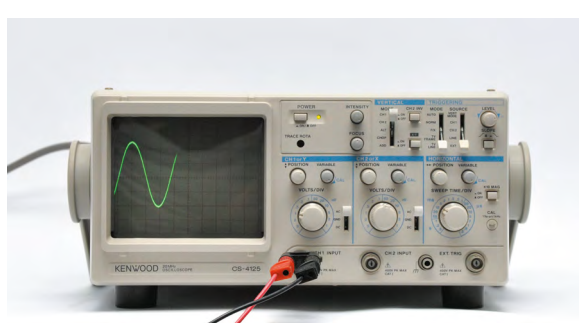


Fig 2.2l *CRO.*

voltmeter 伏特計

Quick check 3

E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 *True or false:*

(a) A battery provides energy so that a current flows in a circuit. (T/F)

(b) The p.d. across a bulb in a circuit is the change in electric PE per electron passing through the bulb. (T/F)

2 Which of the following can be the unit of e.m.f.? Put a tick in the corresponding box(es).

☐ J

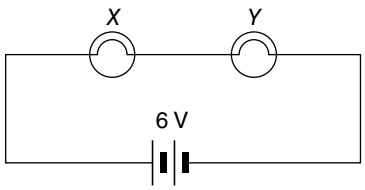
☐ V

☐ J C

☐ J C^{-1}

Learn by Practice 2

1 Two light bulbs are connected to a 6-V battery. The p.d. across bulb *X* is measured to be 4 V by a voltmeter.



(a) Draw in the figure how the voltmeter should be connected to measure the p.d. across bulb *X*. Indicate the positive terminal of the voltmeter.

(b) When 3 C of charges flow through bulb *X*,

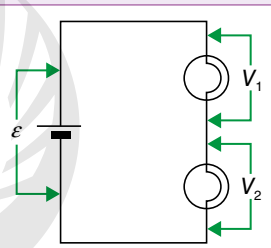
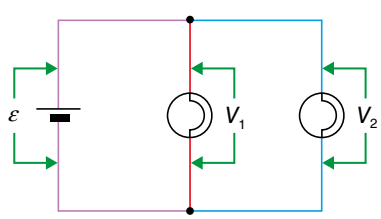
(i) what is the amount of charges that flow through the battery?

(ii) what is the energy imparted by the battery?

4 Voltage and conservation of energy

Energy is conserved in a closed circuit. When charges circulate in the circuit, the energy gained from the source must be equal to the total energy lost in the loads. It can be shown that:

See p.109 for details. ▶

Series connection	Parallel connection
	
$\varepsilon = V_1 + V_2$	$\varepsilon = V_1 = V_2$

The violet, red and blue paths are parallel branches. ▶

Assume no energy loss in the battery and the wires.

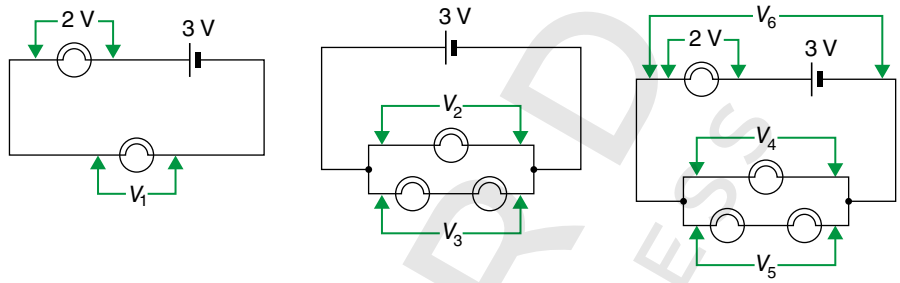
1 In a series circuit, the sum of p.d. across each load is equal to the e.m.f. of the source.

2 In a parallel circuit, the p.d. across each branch is the same.

Example 4 Voltages in circuits

Find the voltages V_1 to V_6 in the following circuits.

(a) (b) (c)



Solution

(a) $V_1 = 3 - 2 = 1$ V

(b) $V_2 = V_3 = 3$ V

(c) $V_4 = V_5 = V_6 = 3 - 2 = 1$ V

Tip
Identify the parallel branches.

5 Cells in series and parallel

For simplicity, all cells are assumed to be identical.

Dry cells can be connected in series or in parallel (Fig 2.2m). What is the difference between these two connections?

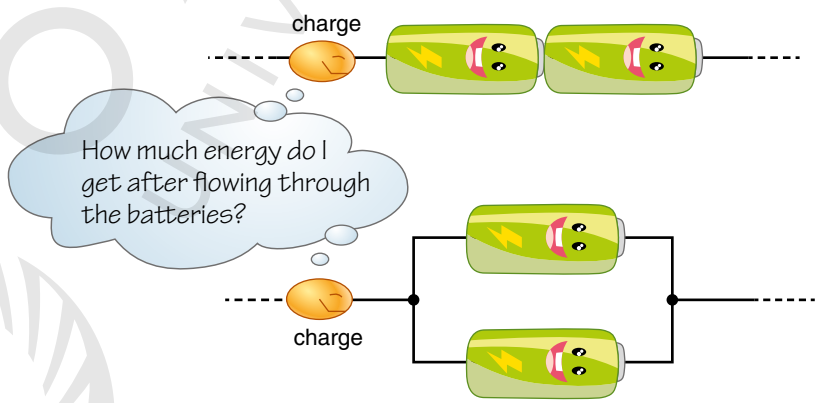


Fig 2.2m Cells in series and in parallel.

Know more

Cells in series

It is very common that cells are connected in series to produce a higher e.m.f.

▸ p.110

All cells are connected in the same direction. If one of the cells is reversed,

total e.m.f. = 1.5 + 1.5 – 1.5
= 1.5 V

In a series connection, a charge flows through all the cells, so the energy it gains is the sum of the energy imparted by all the cells. The total e.m.f. of the cells is therefore the **sum** of all the cells (Fig 2.2n).



Fig 2.2n Series connection of cells.

✎ The maximum current provided by a cell is limited by its power. Hence, identical cells are sometimes connected in parallel to provide a larger output current.

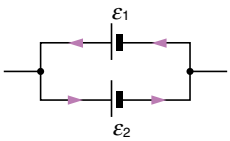
The cells last longer when they are connected in parallel. ▶

In a parallel connection, a charge can only flow through one cell in each round, so the energy it gains is the same as that when there is only one cell. The total e.m.f. of the cells is therefore the same as a single cell (Fig 2.2o).



Fig 2.2o *Parallel connection of batteries.*

You should not connect cells of different e.m.f. in parallel. This is because current will flow in the closed loop formed by the cells until the cells die out.



$(\varepsilon_1 > \varepsilon_2)$

When cells are connected in series, the total e.m.f. is equal to the sum of the e.m.f. of all the cells.
When cells are connected in parallel, the total e.m.f. is equal to the e.m.f. of a single cell.

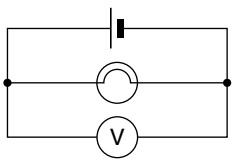
Quick check 4

✎ E-version in OUP Exercise Platform is also available.

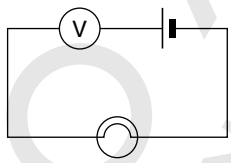
Google Forms | Microsoft Forms

1 In which of the following connections does the voltmeter measure the p.d. across the light bulb? Put a tick in the corresponding box(es).

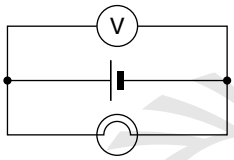
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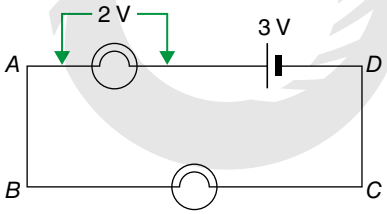
☐



☐



2 Two light bulbs are connected to a battery.



(a) Find the p.d. across *BC*.

(b) Find the p.d. across *AD*.

3 *True or false:*

(a) In a series circuit, the sum of p.d. across each load is equal to the e.m.f. of the source. (T/F)

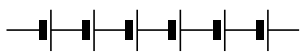
(b) In a parallel circuit, the p.d. across the parallel branches are the same. (T/F)

(c) In a parallel circuit, the p.d. across the parallel branches must be the same as the e.m.f. of the source. (T/F)

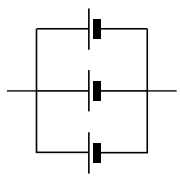
(d) When identical cells are connected in parallel in the same direction, the total e.m.f. is equal to the sum of the e.m.f. of each cell. (T/F)

4 Each cell below has an e.m.f. of 1.5 V. Find the total e.m.f. of each battery.

(a)



(b)



Exercise 2.2

★**1** Two bulbs *X* and *Y* are connected to a 6-V battery (Fig a). If the p.d. across bulb *X* is 2 V, which of the following statements are correct?

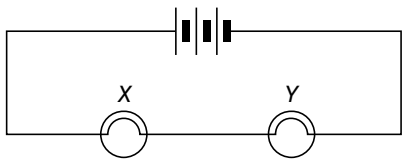


Fig a

(1) When 2 C of charge passes through bulb *X*, 4 J of energy is transferred to the bulb.

(2) When 1 C of charge passes through bulb *Y*, 2 J of energy is transferred to the bulb.

(3) The p.d. across bulb *Y* is 4 V.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★**2** In a closed circuit formed by a battery and a light bulb, the charges

(1) gain energy when they pass through the battery.

(2) slow down when they pass through the light bulb.

(3) speed up when they pass through the battery.

A (1) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★**3**

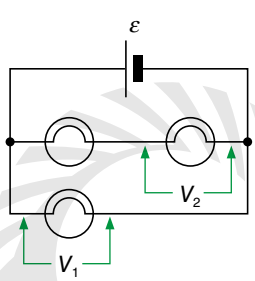


Fig b

Suppose all the bulbs shown in Figure b are identical. Which of the following relationships is correct?

A $V_1 = V_2$

B $V_1 = \varepsilon$

C $V_2 = \varepsilon$

D All of the above

★**4** Bulbs *X* and *Y* are connected to an 8-V battery (Fig c). The current drawn from the battery is 0.6 A.

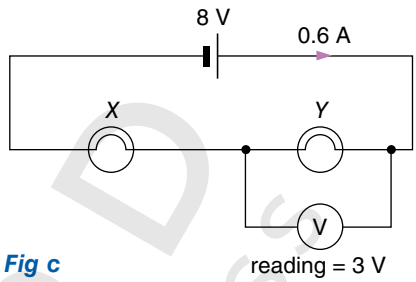


Fig c

(a) Find the p.d. across bulb *X*.

(b) Find the amount of charge passing through bulb *X* in 2 minutes.

(c) Find the energy consumed by bulb *X* in 2 minutes.

★**5** Figure d shows a 1.2-V rechargeable battery rated at 3000 mA h.



Fig d

(a) How many coulombs of charge can be driven to flow through the battery so that its capacity drops from 100% to zero?

(b) Calculate the energy stored in this battery.

(c) If a current of 10 mA is drawn from the battery, how long can the battery last?

★**6** Bulbs *A* and *B* are connected to a battery which consists of two 1.5-V cells (Fig e). Amy wants to measure the p.d. across bulb *B* with a voltmeter.

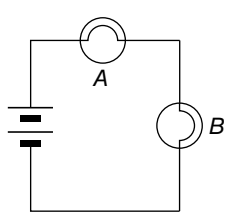


Fig e

(a) Draw the voltmeter with an appropriate connection in the circuit. Indicate the positive terminal of the voltmeter.

(b) Find the e.m.f. of the battery.

(c) If the voltmeter reads 2 V, what is the p.d. across bulb *A*?

(d) Amy claims that a 1.5-V cell provides 1.5 C of charges to the circuit in one second. Comment on Amy's statement.

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2.3 Resistance

1 Definition of resistance

DSE goal
Define resistance.

We must **not** touch any electrical appliances with wet hands; otherwise we might possibly get an electric shock. Why?



Fig 2.3a *Touch an electrical appliance with wet hands may result in an electric shock.*

One of the reason is related to **resistance**. In junior science, you learned that resistance is a measure of the opposition of an object to the flow of current. The resistance R of a conductor is defined as the **ratio** of the **voltage** V across it and the **current** I flowing through it. Its unit is the **ohm** (Ω).

Ω is the Greek letter omega.

1 Ω = 1 V A^{-1}

resistance of conductor (Ω) $R = \dfrac{V}{I}$ **voltage** across conductor (V); **current** through conductor (A)

When the p.d. remains the same, the current is larger if the conductor has a lower resistance.

Know more

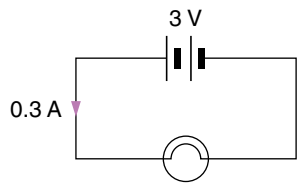
No wet hands!

When the human body is dry, its resistance can be as high as 100 000 Ω. However, if the body is wet, the resistance may become 1000 Ω. Under the same voltage, the current can be 100 times larger if the body is wet.

Moreover, the water may enter the appliance and connect you to the circuit.

Example 5 Resistance of a bulb

A light bulb is connected to a 3-V battery as shown. The current through the bulb is 0.3 A.



(a) Find the resistance R of the bulb.

(b) How will the current change if the bulb is replaced by another one of a higher resistance?

Solution

(a) $R = \dfrac{V}{I} = \dfrac{3}{0.3} = 10\ \Omega$

(b) Decrease

Tip
By $I = \dfrac{V}{R}$, $R \uparrow \Rightarrow I \downarrow$

resistance 電阻　　ohm (Ω) 歐姆

DSE goal

Measure R in simple circuit.

Know more

Multimeter

A multimeter can be used to measure the resistance of an object directly. It can also measure current and voltage. By turning a dial on the multimeter, you can choose the type of measurement and the scale for it.



Video

→ This video shows the use of a multimeter to measure the current through, the voltage across and the resistance of a lamp.

Google Forms

Microsoft Forms

2 Measuring resistance

The resistance of a circuit component, such as a resistor, can be found by using the *voltmeter-ammeter method* (Fig 2.3b). We measure the current I through the resistor and the potential difference V across it. Its resistance can be calculated from $R = \frac{V}{I}$.

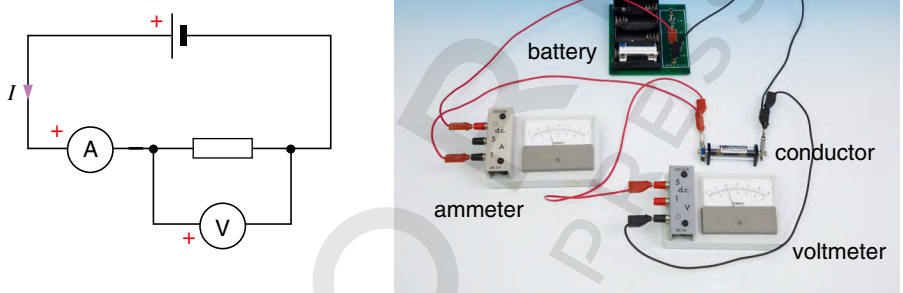


Fig 2.3b The voltmeter-ammeter method.

Alternatively, we can use a *multimeter* to measure resistance directly.

Quick check 5

E-version in OUP Exercise Platform is also available.

1 A conductor is connected to a battery. Complete the following table.

	R / Ω	V / V	I / A
(a)		5	2
(b)	100	3	
(c)	10		0.5

2 Which of the following can be the unit of resistance? Put a tick in the corresponding box(es).

☐ Ω

☐ V A

☐ V A^{-1}

Learn by practice 3

1 With the same p.d. applied, a current of 2 A passes through a bulb P and a current of 3 A passes through another bulb Q. Which bulb has a higher resistance?

2 When a bulb is connected to a 12-V battery, a current of 1.5 A passes through it. Find its resistance.

3 When a resistor is connected to a power supply, a current of 2.4 A passes through it. Its resistance is 50 Ω. Find the p.d. across it.

voltmeter-ammeter method 伏安法　　multimeter 萬用電錶

DSE goal

Describe variation of current with p.d. in metal wires, electrolytes, filament lamps and diodes.

Realize Ohm's law as a special case of resistance behaviour.

DSE exam

18(1A)Q23

Simulation Video

→ This simulation is a 'virtual expt' of Expt 2a.

→ This video demonstrates Expt 2a.

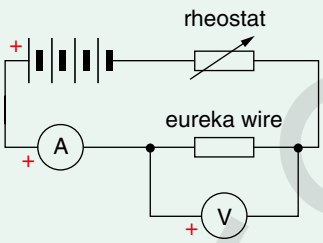
3 Ohm's law

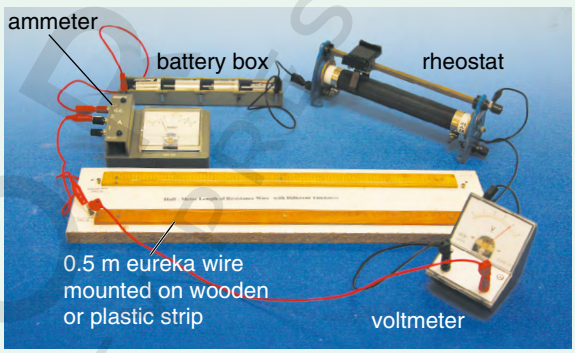
In the following experiment, we will find out whether the resistance of a conductor changes with the p.d. across it.

Experiment 2a / Ohm's law

→ This is a Video Quiz about this experiment.

1 Connect the circuit as shown.





2 Adjust the *rheostat* to vary the current through the *eureka wire*. Take the ammeter reading I and the voltmeter reading V.

3 Repeat step 2 several times. Plot a graph of V against I.

Precaution

1 Close the circuit only when taking readings.

2 The sliding contact of the rheostat should initially be set at the position that gives the largest resistance.

Discussion

Does the resistance of the wire change with the p.d. across it?

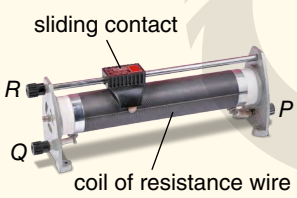
An eureka wire is a copper-nickel alloy. ▶

→ Teaching notes and exercises are available on OUP web.

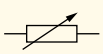
Skill *Experiment*

Rheostat

The resistance of a rheostat can be varied by moving the sliding contact. When terminals P and R are connected, its resistance is the highest by setting the sliding contact at end R.



Its circuit symbol is the same as that of a variable resistor.



In Experiment 2a, we plot a V–I graph for the eureka wire. The graph is a **straight line through the origin** (Fig 2.3c). This shows that the p.d. across the wire is directly proportional to the current through it.

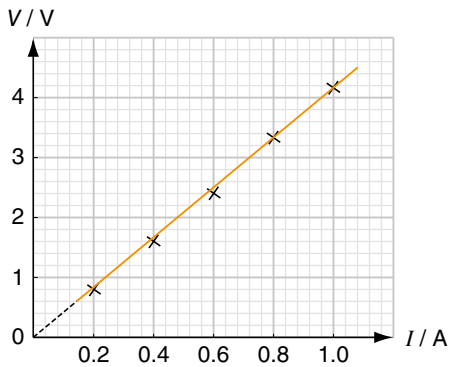


Fig 2.3c *The V–I graph of the eureka wire.*

rheostat 變阻器　　eureka wire 電阻線

This result is known as **Ohm's law**:

> **The p.d. across the ends of a conductor is directly proportional to the current flowing through it ($V \propto I$), provided that temperature and other physical conditions are unchanged.**

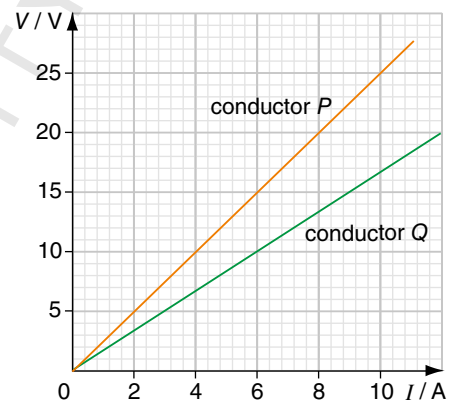
Ohm's law is applicable to many different metals, but not all conductors. Conductors that obey Ohm's law are called **ohmic conductors**. The resistance $\left(R = \frac{V}{I}\right)$ of an ohmic conductor remains the same when different p.d. is applied across it.

Example 6 | *V–I* graphs of ohmic conductors

The *V–I* graphs of two ohmic conductors *P* and *Q* are shown on the right.

(a) Find the resistance of *P*.

(b) Does *Q* have a lower or higher resistance than *P*?



Solution

(a) For conductor *P*, when $V = 25$ V, $I = 10$ A.

Resistance of $P = \frac{V}{I} = \frac{25}{10} = 2.5\ \Omega$

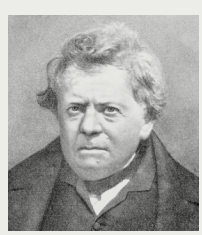
(b) *Q* has a lower resistance than *P*.

For an ohmic conductor, taking any point on the graph to compute $\frac{V}{I}$ will give the same result.

Know more

Georg Ohm (1789–1854)

Georg Ohm was a German physicist. He discovered the Ohm's law. The unit of resistance is named after him.



Caution

Both *I*–*V* and *V*–*I* graphs can be used to describe the relationship between current and p.d. Pay attention to the axes of the graphs.

Ohm's law is a special case of resistance behaviour. Many conductors do not obey Ohm's law and are called **non-ohmic conductors**. Their resistances vary under different p.d. Figure 2.3d shows the *I*–*V* graphs of some ohmic and non-ohmic conductors.

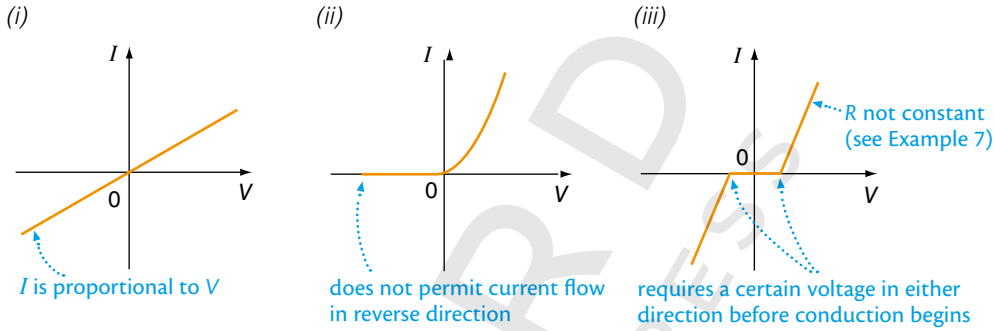


Fig 2.3d *I–V graph of (i) a metal wire (ohmic), (ii) a diode (non-ohmic) and (iii) a kind of electrolyte called dilute sulphuric acid (non-ohmic).*

Ts may explain to Ss that the negative current in Fig 2.3d means the current is flowing the other way round.

Know more

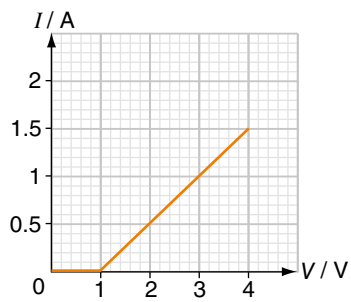
Diode and Electrolyte

A diode permits the flow of current in only one direction. A light-emitting diode (LED) is a diode that emits light.

An electrolyte is a solution with positive and negative ions acting as movable charges.

Example 7 *I–V* graph of a conductor

The p.d. V across a conductor is plotted against the current I passing through it at a constant temperature.



(a) Does the conductor obey Ohm's law? How does the graph indicate this result?

(b) Find the resistance of the conductor when V is equal to (i) 2 V, (ii) 3 V.

Solution

(a) No. The graph is not a straight line passing through the origin.

Caution

Although the graph is a straight line from V = 1 V to 4 V, R is **not** constant.

(b) (i) When V = 2 V, I = 0.5 A.

$$\text{Resistance} = \frac{V}{I} = \frac{2}{0.5} = 4\ \Omega$$

The resistance of the conductor is decreasing from V = 1 V to 4 V.

(ii) When V = 3 V, I = 1 A.

$$\text{Resistance} = \frac{V}{I} = \frac{3}{1} = 3\ \Omega$$

Right and wrong

✗ R is obtained by finding the slope of a *V*–*I* graph or the reciprocal of slope of an *I*–*V* graph.

✓ By definition, R is the ratio of V and I at a **point** on the graph.

diode 二極管　　electrolyte 電解質

Note that by definition, R always equals $\frac{V}{I}$, no matter whether the conductor is ohmic or not. Table 2.3a compares ohmic and non-ohmic conductors.

Ohmic conductor	Non-ohmic conductor
$R = \frac{V}{I}$ = constant	$R = \frac{V}{I} \neq$ constant
obey Ohm's law	not obey Ohm's law
V–I graph is a straight line through origin	V–I graph is **not** a straight line through origin

Table 2.3a *Comparing ohmic and non-ohmic conductors.*

Quick check 6

E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

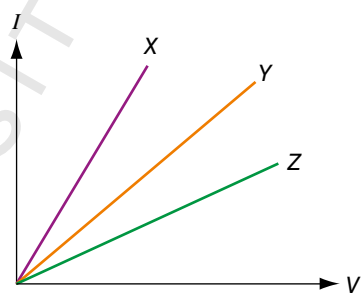
1 The p.d. across conductor X is varied. The temperature and other physical conditions of X are unchanged. Determine whether the following statements about X are true.

(a) If it is ohmic, its resistance is independent of the current through it. (T/F)

(b) Its resistance must be equal to the slope of its V–I graph. (T/F)

(c) The equation $R = \frac{V}{I}$ is applicable only if X is ohmic. (T/F)

2 The I–V graphs of three conductors X, Y and Z are as shown. Which one has the highest resistance?



Learn by practice 4

1 The V–I graph of a conductor X is shown below.

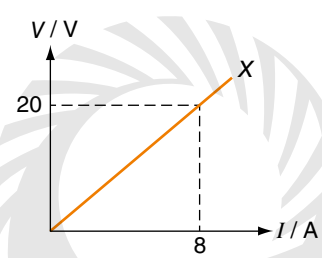


Fig a

(a) Find the resistance of X.

(b) Another ohmic conductor Y has a higher resistance than X.

(i) Sketch a possible V–I graph of Y in the figure.

(ii) Sketch the shapes of the I–V graphs of X and Y.

2 The I–V graph of a conductor Z is shown below.

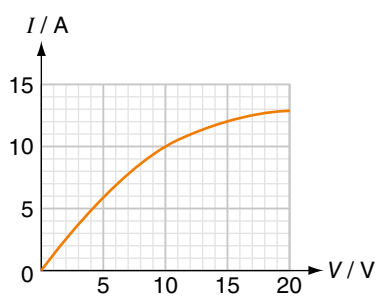


Fig b

(a) Find the resistance of Z when the p.d. across it is (i) 10 V, (ii) 20 V.

(b) Describe how the resistance of Z changes with the p.d. across it.

DSE goal

Determine factors affecting resistance of wire.

Define resistivity of wire.

Describe effect of temperature on resistance of metals and semiconductors.

DSE exam

13(1B)Q10(a), 17(1B)Q8, 18(1A)Q25, 21(1A)Q23, 22(1B)Q8(b), 23(1B)Q7(a)

Simulation Video

→ This simulation is a 'virtual expt' of Expt 2b.

→ This video demonstrates Expt 2b.

Be careful! Do not touch the hot wire! Disconnect the circuit once the readings are taken.

→ This is a Video Quiz about this experiment.

4 Factors affecting resistance

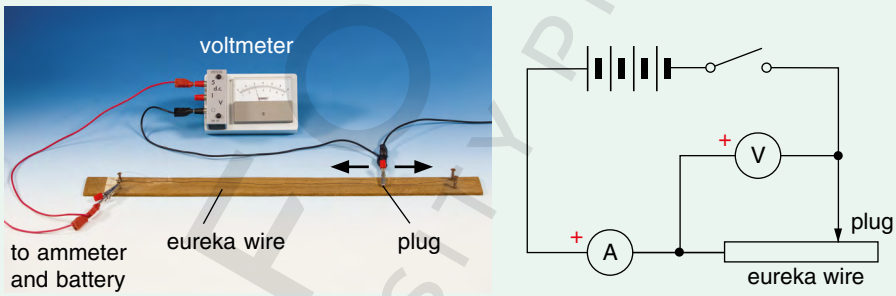
The resistance of a wire depends on several factors. We shall study the factors below.

a Length, thickness and resistivity

Experiment 2b / Effect of length and thickness on resistance of metal wire

Part I Length

1 Connect the circuit as shown.



2 Slide the plug along the eureka wire to change the length of wire connected in the circuit. Take the ammeter readings I and voltmeter readings V accordingly. Hence, calculate the resistance R.

Part II Thickness

1 Keep using the set-up in **Part I**. Measure the current and the p.d. across the whole wire and calculate its resistance.

2 Repeat step 1 with a thicker wire of the same material and length.

Discussion

1 Which one has a lower resistance, a long or a short segment of wire?

2 Which one has a lower resistance, the thick or the thin wire?

Experiment 2b shows that (Fig 2.3e):

1 the longer (larger l) the wire, the higher its resistance R;

2 the thicker (larger cross-sectional area A) the wire, the lower its resistance.

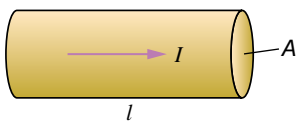


Fig 2.3e *Length l and cross-sectional area A of a wire.*

Further studies show that for wires of the same material,

1 R is directly proportional to l, i.e. $R \propto l$;

2 R is inversely proportional to A, i.e. $R \propto \frac{1}{A}$.

The three quantities are therefore related as $R \propto \frac{l}{A}$, or

$$R = \rho \frac{l}{A}$$

resistance (Ω), length (m), resistivity (Ω m), cross sectional area (m^2)

where ρ is the proportionality constant called **resistivity**.

Rearranging the equation, we have $\rho = \frac{RA}{l}$. This gives the definition of the resistivity of a material. Its unit is Ω m.

Unit of ρ
$= \frac{\text{unit of } (R \times A)}{\text{unit of } l}$
$= \frac{\Omega \times m^2}{m}$
$= \Omega$ m

Resistivity is a measure of how strongly a material opposes the flow of current. It **depends on the type of material but not the dimension** (length and thickness) of a conductor. A good conductor like copper has a low resistivity. Figure 2.3f gives the resistivities of some materials.

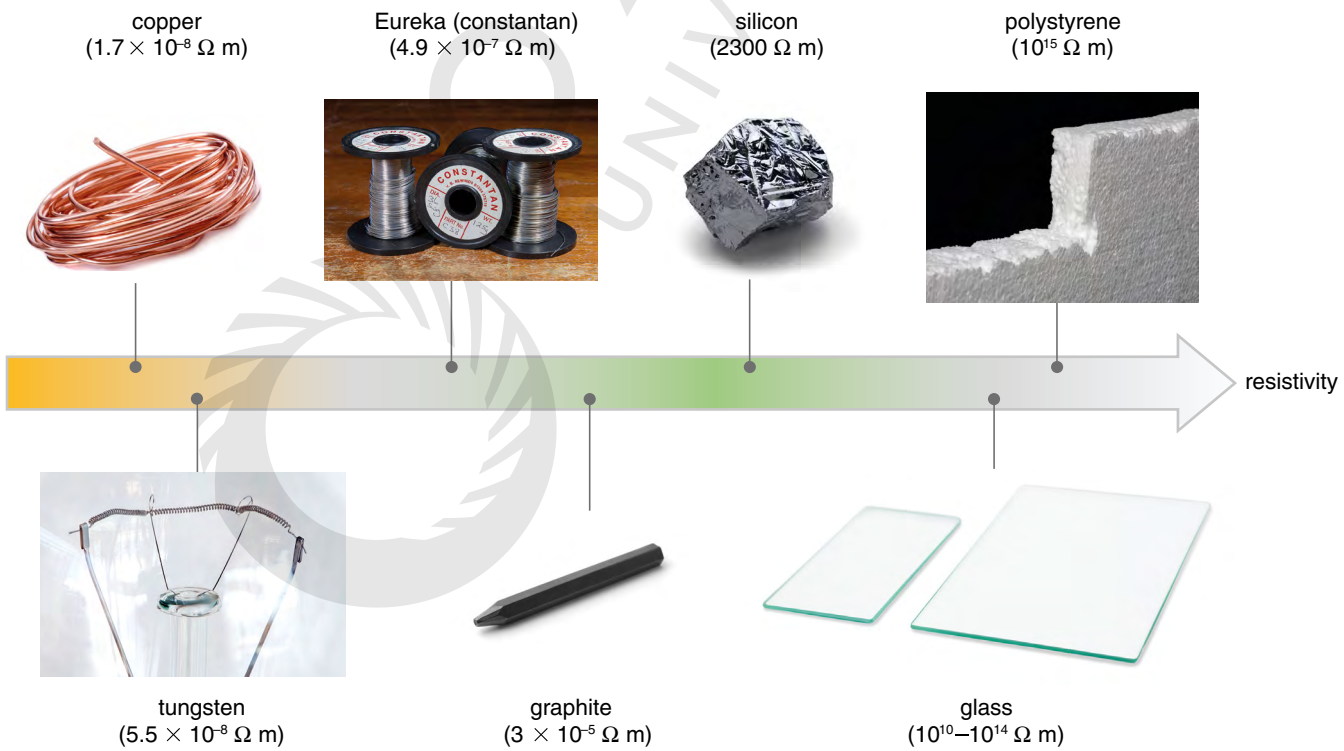


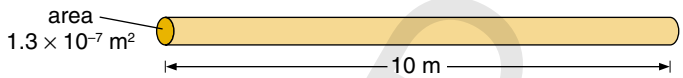
Fig 2.3f *Resistivity of different materials at 20 °C.*

resistivity 電阻率

Example 8 Resistivity and resistance

A copper wire is 10 m long. Its cross-sectional area is 1.3×10^{-7} m^2. Its resistance is 1.34 Ω.

Copper wires have very low resistance and are used as connecting wires in circuits. We usually assume the resistance of connecting wires as zero.



(a) Find the resistivity of the copper wire.

(b) The length of the wire is reduced by half. How do its resistance and resistivity change?

Solution

(a) $\rho = \dfrac{RA}{l} = \dfrac{(1.34)(1.3 \times 10^{-7})}{10} = 1.74 \times 10^{-8}\ \Omega\ \text{m}$

(b) $R = \rho \dfrac{l}{A} \propto l$ (for constant ρ and A)

$\therefore$ When l is halved, R is also halved.

The resistivity remains unchanged.

Tip

Resistivity depends on the material but not the dimension.

b Temperature

You may have noticed that the temperature is specified when listing the resistivities in Figure 2.3f (p.80). This is because the resistivity, hence the resistance, of a material changes with temperature.

Experiment 2c Effect of temperature on resistance of metal wire

Video

→ This video demonstrates Expt 2c.

1 Wind a 1-m metal wire into a coil.

2 Connect the coil to a multimeter and measure its resistance (Fig a).

3 Heat the coil and measure its resistance again (Fig b).

Do **not** move any contacts while measuring the resistance of the wire. A poor contact greatly affects your results.

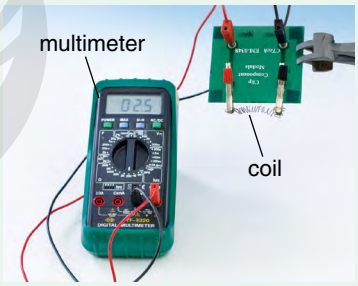


Fig a



Fig b

Be careful! Do not touch the hot wire!

→ This is a Video Quiz about another experiment investigating the effect of temperature on the resistance of a metal wire.

Discussion

How does the resistance of the wire change with its temperature?

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✎ Tungsten has a melting point of 3422 °C, which is the highest among all non-alloyed metals. Tungsten filament bulbs are filled with inert gas or with a vacuum inside to prevent oxidization of the filament.

✎ Remind Ss that although the graph in Fig 2.3g is a curve, we cannot draw a conclusion that tungsten is a non-ohmic conductor. This is because one of the conditions of Ohm's law (temperature unchanged) is not satisfied in this case.

In the experiment, the resistance of the wire increases with temperature. The tungsten filament in a bulb (Fig 2.3g) behaves similarly. When the bulb is switched on, the filament heats up. **The larger the current, the hotter the filament and the higher the resistance.** The resistance can increase by more than 10 times.



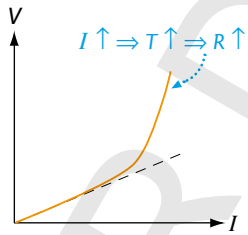


Fig 2.3g *The filament in a bulb and its V–I graph.*

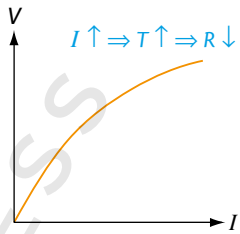


Fig 2.3h *The V–I graph of a typical semiconductor.*

Metals can be used in resistance thermometers while semiconductors can be used in thermistor thermometers because their resistances vary with temperature.

▶ In general, the resistance of a metal increases with temperature, while the resistance of a *semiconductor* decreases as temperature increases (Fig 2.3h).

The resistance of a wire depends on its length, cross-sectional area, resistivity (material) and temperature.

Google Forms Microsoft Forms

Quick check 7

✎ E-version in OUP Exercise Platform is also available.

1 The unit of resistivity is

A Ω B Ω m

C Ω m^{-1} D Ω m^{-2}

2 *True or false:*

(a) A longer wire must have a higher resistance than a shorter wire. (T/F)

(b) A longer copper wire has a higher resistivity than a shorter copper wire of the same thickness. (T/F)

(c) In general, the resistance of a metal increases with temperature. (T/F)

3 Complete the following table for different conductors.

	R / Ω	ρ / Ω m	l / m	A / m^2
X		100×10^{-8}	20	4×10^{-5}
Y	100		1	2×10^{-3}
Z	2×10^{-3}	3×10^{-5}		6×10^{-4}

Learn by practice 5

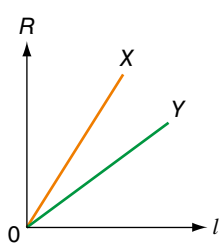
1 A piece of metal wire is 15 cm long and has a diameter of 1.6 mm. If the resistance of the metal wire is 3 Ω, what is its resistivity?

2 How is the resistance and the resistivity of a steel wire affected if

(a) its diameter is doubled?

(b) its temperature is increased significantly?

3 The graph shows the resistance R of two copper wires, X and Y, against their lengths l. Which wire is thicker?



semiconductor 半導體

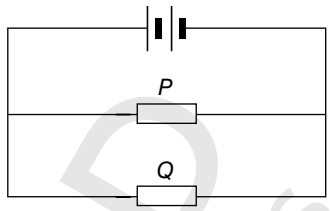
☆ HKDSE 2018 Paper 1A Q25

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Resistances of two wires

Iron wires P and Q have the same length. When they are connected to a battery as shown, the current passing through P is twice that passing through Q. If the radius of P is r, what is the radius of Q?



A $\frac{r}{2}$ B $\frac{r}{\sqrt{2}}$

C $\sqrt{2}r$ D $2r$

Solution

By $R = \frac{V}{I}$ and $R = \frac{\rho l}{A} = \frac{\rho l}{\pi r^2}$,

$\frac{V}{I} = \frac{\rho l}{\pi r^2}$ — Look for the relationship of the quantities given.

$r^2 \propto I$ ($\because P$ and Q have the same V, ρ and l) — P and Q are connected in parallel and have the same V.

$\therefore \frac{r_Q^2}{r_P^2} = \frac{I_Q}{I_P}$

$\frac{r_Q^2}{r^2} = \frac{I}{2I}$

$r_Q = \frac{r}{\sqrt{2}}$

$\therefore$ The answer is B.

Revision 2 Q19 (p.104)

Exercise 2.3

1

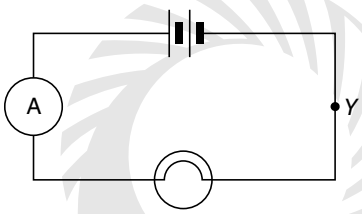


Fig a

Which of the following changes would result in a larger reading of the ammeter in Figure a?

(1) Use a bulb with a higher resistance.

(2) Use a battery of a higher e.m.f.

(3) Move the ammeter to position Y.

A (1) only B (2) only

C (1) and (2) only D (2) and (3) only

***2**

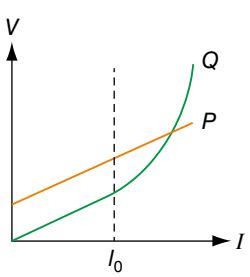


Fig b

P and Q are two conductors. Their V–I graphs are shown in Figure b. Which of the following statements is/are correct?

(1) P obeys Ohm's law.

(2) Q obeys Ohm's law when $I < I_0$.

(3) P has a higher resistance than Q when $I < I_0$.

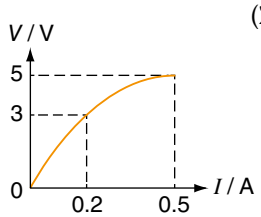
A (2) only B (1) and (2) only

C (1) and (3) only D (2) and (3) only

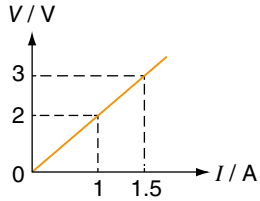
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★**3** The following graphs show the relationship of V and I of different conductors. Which graph(s) show(s) that the resistance of the conductor increases when V increases?

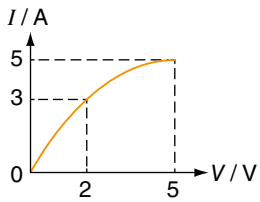
(1)



(2)



(3)



A (1) only B (3) only

C (1) and (2) only D (2) and (3) only

★**4** An ohmic conductor has a resistance of 3 Ω. In Figure c, draw its V–I graph, provided that its temperature remains unchanged.

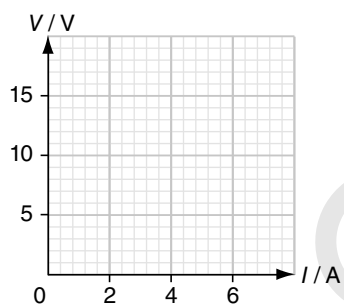


Fig c

★**5** The I–V graphs of two conductors X and Y are as shown (Fig d). The conductors remain at the same temperature during the whole process.

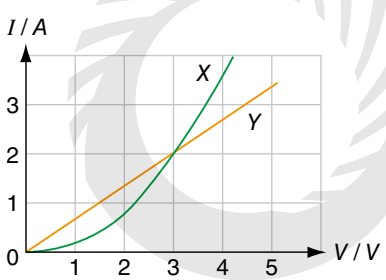


Fig d

(a) How do the resistances of X and Y change when V increases from 0 to 3 V?

(b) Conductor Z is an ohmic conductor. Its resistance is lower than Y. Sketch its I–V graph in Figure d.

★**6** A copper wire is connected in a circuit and a current passes through it. The wire has a length of 15 cm and a radius of 0.2 mm. The resistivity of copper is 1.7×10^{-8} Ω m.

(a) Find the resistance of the wire.

(b) The copper wire is then replaced by a zinc wire of the same length and radius. Zinc has a higher resistivity than copper. How will the current passing through the wire change? Explain your answer by assuming the p.d. across the wire remains the same.

★**7** Figure e shows the I–V graph of a diode at a constant temperature.

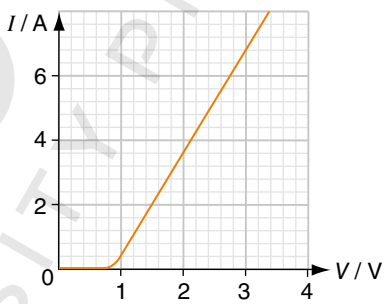


Fig e

(a) What is the minimum applied voltage needed for a current to pass through the diode?

(b) Does this diode obey Ohm's law?

(c) Find the resistance of the diode when $V = 2$ V.

(d) How does the resistance of the diode change when V increases from 2 V to 3 V?

★**8** The V–I graph of a cubic semiconductor X is shown in Figure f. X becomes hotter when V increases.

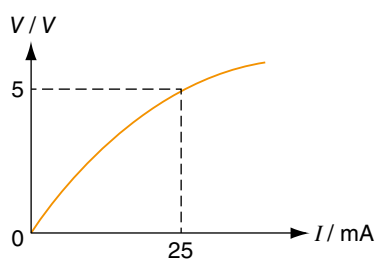


Fig f

(a) How does the resistance of X change with temperature?

(b) Can the equation $R = \frac{V}{I}$ be used to find the resistance of X? Explain briefly.

(c) The resistivity of X is 0.4 Ω m when I is 25 mA. Find the length of X.

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2.4 Resistor networks

Current flows when a resistor is connected to a battery. If one more resistor is connected to the circuit in different ways (Fig 2.4a), how will the current change?

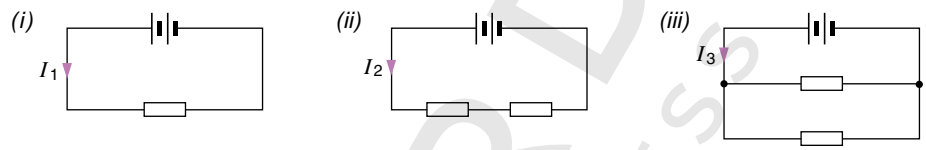


Fig 2.4a *Circuits with (i) one resistor, (ii) two resistors in series and (iii) two resistors in parallel.*

Experiments show that $I_3 > I_1 > I_2$. This can be explained by considering the resistance of the circuit.

Flipped classroom

Watch a video and answer the questions.

1 Equivalent resistance

There may be many resistors in a circuit. When we analyze such circuits, it is useful to regard a resistor network as a single resistor causing the same effect on the circuit. This single resistor is said to have **equivalent resistance** R_{eq} as the resistor network (Fig 2.4b).

The concept of equivalent resistance is very useful, especially when we study complicated resistor networks.

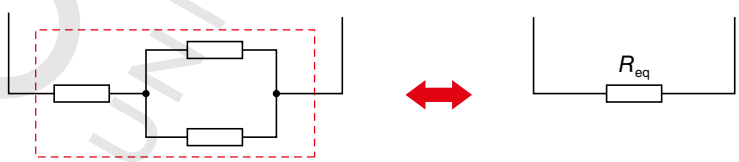


Fig 2.4b *A resistor with resistance R_{eq} causes the same effect as the resistor network.*

DSE goal

Derive $R = R_1 + R_2 + \ldots$ for resistors in series.

DSE exam

23(1A)Q25

Video

→ The video shows an experiment of finding the equivalent resistance of two resistors connected in series.

2 Resistor in series

We will now derive the equivalent resistance of two resistors connected in series (Fig 2.4c).

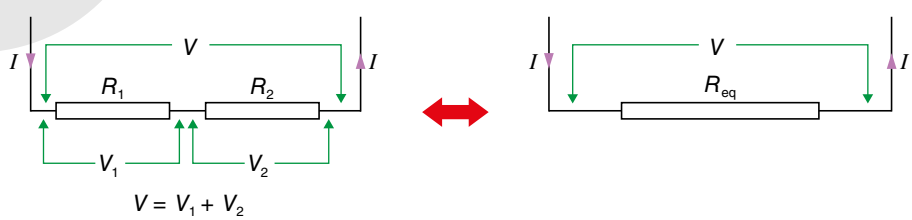


Fig 2.4c *Two resistors connected in series and a resistor equivalent to them.*

equivalent resistance 等效電阻

Know more

Avoid shocks due to static electricity

When you touch a charged object, a current flows through you. You feel an electric shock if the current is large enough. If you touch the charged object with a device of high resistance, you will be connected in series with the device, so the total resistance of you and the device will be much higher and the current will be much smaller. This prevents you from feeling an electric shock.

Video

→ The video shows a DIY anti static electricity stick.

The sum of the p.d. across each of the resistors R_1 and R_2 is equal to the total p.d. across the two resistors:

$$V = V_1 + V_2 \quad \ldots\ldots (1)$$

As R_1 and R_2 are in series, the same current I passes through them.

Now, suppose R_1 and R_2 are replaced with a single resistor R_{eq} such that the values of I and V remain unchanged. By $V = IR$, (1) becomes

$$IR_{eq} = IR_1 + IR_2 = I(R_1 + R_2)$$

$$R_{eq} = R_1 + R_2$$

In general, if n resistors are connected in series:

equivalent resistance (Ω) — resistance of resistors in series (Ω)

$$R_{eq} = R_1 + R_2 + \ldots + R_n$$

R_{eq} is always **larger** than the resistance of any single resistor in the series combination. The effect is like joining two short lengths of resistance wire to make a longer length (Fig 2.4d).

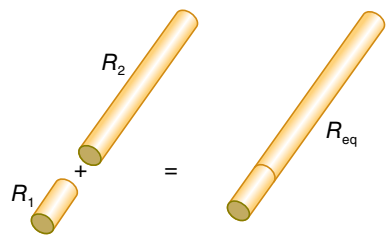


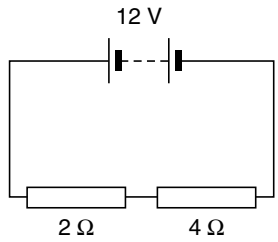
Fig 2.4d Joining resistance wires end on end gives a higher resistance.

For resistors connected in series, if one resistor has a resistance R **much larger** than the others, $R_{eq} \approx R$.

See Check point 8 Q1(a) (p.87). ▶

Example 9 Series connection

Two resistors are connected in series to a battery as shown.



(a) What is the equivalent resistance of the two resistors?

(b) What is the p.d. across each resistor?

Solution

(a) $R_{eq} = 2 + 4 = 6\ \Omega$

(b) Current I through resistors $= \dfrac{V}{R_{eq}} = \dfrac{12}{6} = 2$ A

P.d. across 2-Ω resistor $= IR = 2 \times 2 = 4$ V

P.d. across 4-Ω resistor $= IR = 2 \times 4 = 8$ V

Example 9 shows that the resistor with a **higher resistance** takes up a **greater proportion of the total p.d.** across the series combination. When two resistors are connected in series (Fig 2.4e),

Try Exercise 2.4 Q2 (p.93) to derive these equations. ▶

$$V_1 = \frac{R_1}{R_1 + R_2} \times V$$

$$V_2 = \frac{R_2}{R_1 + R_2} \times V$$

$$V_1 : V_2 = R_1 : R_2$$

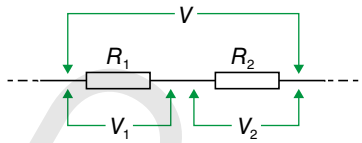


Fig 2.4e *P.d. across two resistors in series.*

Google Forms Microsoft Forms

Quick check 8

E-version in OUP Exercise Platform is also available.

1 Three resistors are connected in series. What is their equivalent resistance?

	R_1 / Ω	R_2 / Ω	R_3 / Ω	R_{eq} / Ω
(a)	1000	2	3	
(b)	2	4		12

(For Q2–3.) Three resistors are connected as shown (Fig a).

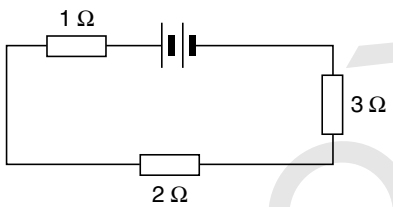


Fig a

2 The equivalent resistance of the three resistors must be

A larger than 3 Ω.

B between 1 Ω and 3 Ω.

C smaller than 1 Ω.

3 The p.d. across which resistor is the largest?

A 1-Ω resistor

B 2-Ω resistor

C 3-Ω resistor

D Cannot be determined.

4 Two resistors are connected to a 9-V battery (Fig b).

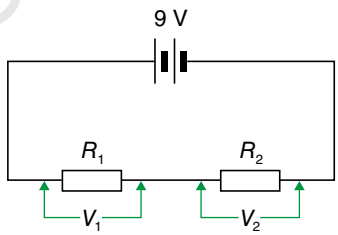


Fig b

Complete the table.

	R_1 / Ω	R_2 / Ω	V_1 / V	V_2 / V
(a)	2	2		
(b)	10		3	
(c)		20	4	

Learn by practice 6

1 When n 8-Ω resistors are connected in series, the equivalent resistance is 96 Ω. Find n.

2 A 6-Ω resistor and a 15-Ω resistor are connected in series. If the voltage across the 6-Ω resistor is 3 V, what is the voltage across the 15-Ω resistor?

3 A 20-Ω resistor and a 30-Ω resistor are connected in series to a 6-V battery. Find the voltage across the 30-Ω resistor.

4 A 5-Ω resistor and a 10-Ω resistor are connected in series to a battery. The voltage across the 10-Ω resistor is 2 V. Find the e.m.f. of the battery.

DSE goal

Derive $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$ for resistors in parallel.

Compare series and parallel circuits in terms of p.d. and current.

DSE exam

16(1A)Q25

Video

→ The video shows an experiment of finding the equivalent resistance of two resistors connected in parallel.

3 Resistors in parallel

The equivalent resistance of two resistors connected in parallel can be found as follows (Fig 2.4f). The sum of the currents flowing through resistors R_1 and R_2 is equal to the current in the main circuit, i.e.

$$I = I_1 + I_2 \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

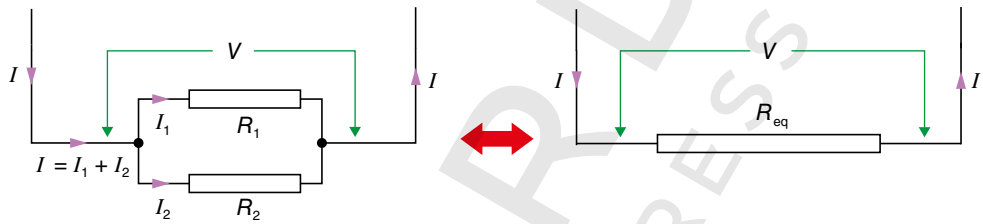


Fig 2.4f *Two resistors connected in parallel and a resistor equivalent to them.*

As R_1 and R_2 are parallel, the p.d. across each resistor is V. Suppose R_1 and R_2 are replaced with a single resistor R_{eq} such that the values of I and V remain unchanged. By $I = \frac{V}{R}$, (2) becomes:

$$\frac{V}{R_{eq}} = \frac{V}{R_1} + \frac{V}{R_2} = V\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

$$\frac{\mathbf{1}}{\boldsymbol{R_{eq}}} = \frac{\mathbf{1}}{\boldsymbol{R_1}} + \frac{\mathbf{1}}{\boldsymbol{R_2}} \quad \textbf{or} \quad \boldsymbol{R_{eq}} = \frac{\boldsymbol{R_1 R_2}}{\boldsymbol{R_1 + R_2}}$$

equivalent resistance (Ω) — resistance of resistors in parallel (Ω)

In general, if n resistors are connected in parallel:

$$\frac{\mathbf{1}}{\boldsymbol{R_{eq}}} = \frac{\mathbf{1}}{\boldsymbol{R_1}} + \frac{\mathbf{1}}{\boldsymbol{R_2}} + \ldots + \frac{\mathbf{1}}{\boldsymbol{R_n}}$$

R_{eq} is always **smaller** than the resistance of any single resistor in the parallel combination. The effect is the same as putting pieces of resistance wire side by side. They behave like a wider (larger cross-sectional area) piece of wire (Fig 2.4g).

$R = \rho \frac{l}{A} \propto \frac{1}{A}$

✎ This is a good analogy to explain why resistors connected in parallel result in a lower equivalent resistance.

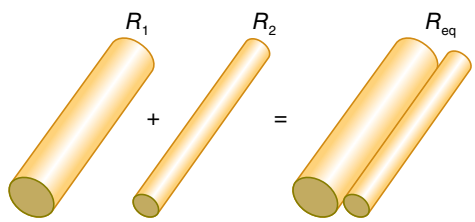


Fig 2.4g *Bundling resistance wires together gives a lower resistance.*

If one resistor has a resistance R **much smaller** than the others,

See Quick check 9 Q1(b) (p.90).

$$R_{eq} \approx R$$

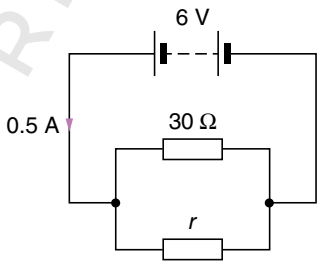
If n **identical** resistors R are connected in parallel,

$$\frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{R} + \ldots + \frac{1}{R} = \frac{n}{R}$$

$$R_{eq} = \frac{R}{n}$$

Example 10 Parallel connection

Two resistors are connected in parallel to a battery. The current drawn from the battery is 0.5 A.



(a) What is the equivalent resistance of the two resistors?

(b) Find the resistance of resistor r.

(c) Find the current passing through each resistor.

Solution

(a) $R_{eq} = \frac{V}{I} = \frac{6}{0.5} = 12\ \Omega$

(b) $\frac{1}{r} + \frac{1}{30} = \frac{1}{R_{eq}}$

$$= \frac{1}{12}$$

$$r = 20\ \Omega$$

(c) I through 30-Ω resistor $= \frac{V}{R} = \frac{6}{30} = 0.2$ A

Alternatively, $I = 0.5 - 0.2 = 0.3$ A

I through resistor $r = \frac{V}{R} = \frac{6}{20} = 0.3$ A

The result shows that the resistor with a **lower resistance** takes up a **greater proportion of the total current** in the parallel combination. When two resistors are connected in parallel (Fig 2.4h),

Try Exercise 2.4 Q3 (p.93) to derive these equations.

$$I_1 = \frac{R_2}{R_1 + R_2} \times I$$

$$I_2 = \frac{R_1}{R_1 + R_2} \times I$$

$$I_1 : I_2 = R_2 : R_1$$

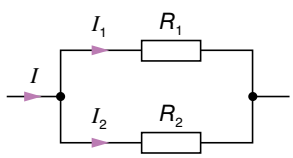


Fig 2.4h Currents through two resistors in parallel.

Table 2.4a compares resistors connected in series and in parallel in terms of p.d., current and equivalent resistance.

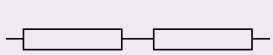
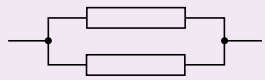
	In series 	In parallel 
***V* across resistor**	larger V across the resistor with a higher resistance	same for each resistor
***I* through resistor**	same for each resistor	larger I through the resistor with a lower resistance
Equivalent resistance R_{eq}	larger than any single resistor	smaller than any single resistor

Table 2.4a *Comparing resistors in series and in parallel.*

Quick check 9

E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

1 Three resistors are connected in parallel. Complete the following table.

	R_1 / Ω	R_2 / Ω	R_3 / Ω	R_{eq} / Ω
(a)	3	3	3	
(b)	1000	1	2000	

(For Q2–3.) Consider the circuit in Figure a.

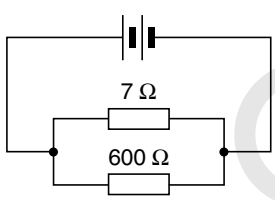


Fig a

2 The equivalent resistance of the resistors is

A smaller than 7 Ω.

B between 7 Ω and 600 Ω.

C greater than 600 Ω.

3 The current through which resistor is larger?

A 7-Ω resistor

B 600-Ω resistor

4 Two resistors are connected to a battery (Fig b).

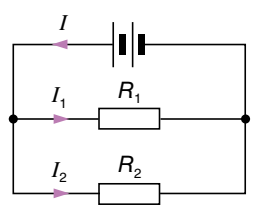


Fig b

Complete the table.

	R_1 / Ω	R_2 / Ω	I / A	I_1 / A	I_2 / A
(a)	2	2	3		
(b)	30			0.2	0.4

Learn by practice 7

1 Find the equivalent resistance when ten 5-Ω resistors are connected in parallel.

2 A 3-Ω resistor and a 5-Ω resistor are connected in parallel to a battery. The current passing through the 3-Ω resistor is 0.6 A. What is the current through the 5-Ω resistor?

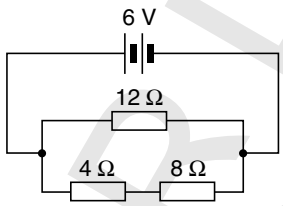
3 A 10-Ω resistor and a 15-Ω resistor are connected in parallel to a battery. The current through the 10-Ω resistor is 0.3 A. Find the current drawn from the battery.

Cross-link
More complicated cases will be discussed in Chapter 3.

A circuit may have some resistors connected in series and some in parallel. We will look at some simple cases in this part.

Example 11 Resistor network

Three resistors are connected to a battery.

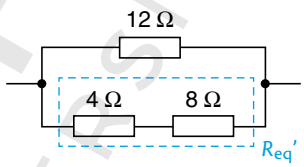


(a) Find the equivalent resistance of the circuit.

(b) Find the p.d. across and the current through each resistor.

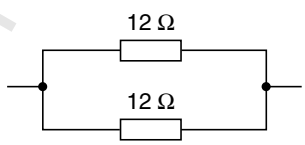
Solution

(a) Consider the 4-Ω and 8-Ω resistor.



$R_{eq}' = 4 + 8 = 12\ \Omega$

Consider the whole network.



$R_{eq} = \frac{12}{2} = 6\ \Omega$

Tip
The branch with the 4-Ω and 8-Ω resistors, and the branch with the 12-Ω resistor are parallel to the battery, so the p.d. across each branch is 6 V.

(b) P.d. across 12-Ω resistor = 6 V

Apply $I = \frac{V}{R}$.

I through 12-Ω resistor $= \frac{6}{12} = 0.5$ A

I through 4-Ω resistor = I through 8-Ω resistor $= \frac{6}{4 + 8} = 0.5$ A

Apply $V = IR$.

P.d. across 4-Ω resistor $= 0.5 \times 4 = 2$ V

P.d. across 8-Ω resistor $= 0.5 \times 8 = 4$ V

Alternative method:
p.d. across 4-Ω resistor
$= \frac{4}{4 + 8} \times 6$
$= 2$ V

Skill *Problem-solving*

Resistor network

Finding equivalent resistance:

① Choose a partial network which is in series or parallel connection and calculate R_{eq} for it.

② Replace that partial network with a single resistor of resistance R_{eq}.

③ Repeat the steps for the other parts.

→ Teaching notes and exercises are available on OUP web.

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Writing practice

Two identical resistors r are connected in series to a battery of e.m.f. ε. The current drawn from the battery is I (Fig a).

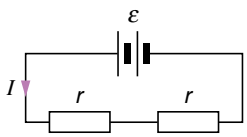


Fig a

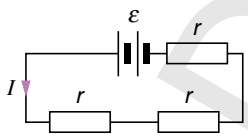


Fig b

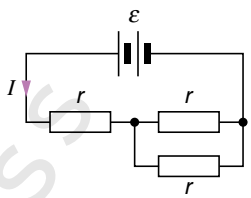


Fig c

Question 1

Explain how I will change if one more resistor r is connected as shown in Figure b. (2 marks)

Answer

The equivalent resistance of the circuit will increase. 1A

By $I = \frac{V}{R}$, I will decrease. 1A

Question 2

Explain how I will change if one more resistor r is connected as shown in Figure c. (2 marks)

Answer

The equivalent resistance of the circuit will decrease. 1A

By $I = \frac{V}{R}$, I will increase. 1A

Tip

$R_{eq} = r + \frac{r}{2} = 1.5r$

Practise yourself

Two identical resistors r are connected in parallel to a battery of e.m.f. ε. The current drawn from the battery is I (Fig d).

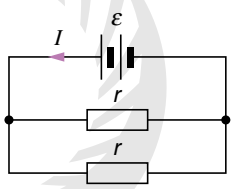


Fig d

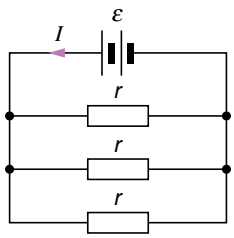


Fig e

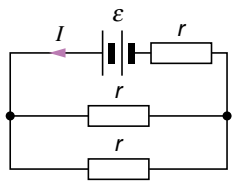


Fig f

1 Explain how I will change if one more resistor r is connected as shown in Figure e. (2 marks)

2 Explain how I will change if one more resistor r is connected as shown in Figure f. (2 marks)

Quick check 10

E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 Three resistors R_1, R_2 and R_3 are connected as shown in Figure a. The equivalent resistance across A and B is R_{eq}. Complete the table below.

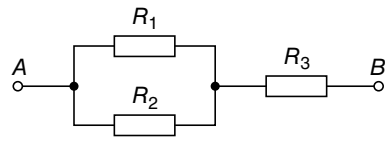


Fig a

	R_1 / Ω	R_2 / Ω	R_3 / Ω	R_{eq} / Ω
(a)	5	15	6	
(b)	1		5	5.75

2 Three resistors R_1, R_2 and R_3 are connected as shown in Figure b. The equivalent resistance across A and B is R_{eq}. Complete the table below.

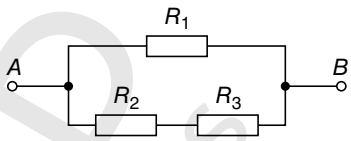


Fig b

	R_1 / Ω	R_2 / Ω	R_3 / Ω	R_{eq} / Ω
(a)	10	10	20	
(b)	8		5	4

Exercise 2.4

★**1** Two resistors R_1 and R_2 are connected to a battery. Which of the following statements must be correct?

(1) The equivalent resistance of the resistors is larger than R_1.

(2) The sum of the p.d. across each resistor is equal to the e.m.f. of the battery.

(3) The total energy consumed by the resistors is equal to the energy imparted by the battery.

A (2) only

B (3) only

C (1) and (2) only

D (1), (2) and (3)

★**2** Two resistors are connected in series in a circuit (Fig a).

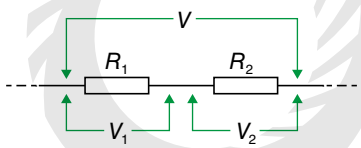


Fig a

Find the following in terms of V, R_1 and R_2.

(a) Equivalent resistance of R_1 and R_2

(b) Current passing through R_1

(c) V_1 and V_2

(d) $V_1 : V_2$

★**3** Two resistors are connected in parallel in a circuit (Fig b).

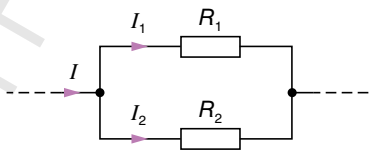


Fig b

Find the following in terms of I, R_1 and R_2.

(a) Equivalent resistance of R_1 and R_2

(b) P.d. across R_1

(c) I_1 and I_2

(d) $I_1 : I_2$

★**4** There are 3 resistors of resistance 1 Ω, 10 Ω and 100 Ω.

(a) Two of the resistors are connected in series to a battery such that the equivalent resistance R_{eq} is the largest.

(i) Which two resistors are connected?

(ii) Which resistor has a larger current flowing through it?

(iii) Which resistor has a higher p.d. across it?

(b) Two of the resistors are connected in parallel to a battery such that the equivalent resistance R_{eq} is the smallest.

(i) Which two resistors are connected?

(ii) Which resistor has a larger current flowing through it?

(iii) Which resistor has a higher p.d. across it?

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2.5 Electrical power

Do you know which light bulb shown below is brighter if they are connected together to a circuit?



Fig 2.5a *Packages of two light bulbs.*

You may choose the 60-W bulb. In fact, the answer depends on how the bulbs are connected! Let us look into this step by step.

DSE goal

Apply $P = VI$ to solve problems.

DSE exam

13(1A)Q30, 16(1A)Q26, 20(1A)Q22, 21(1A)Q27, 22(1B)Q8(c)

1 Definition of power

The brightness of a light bulb depends on its **electrical power**. Recall that power is the rate at which energy is converted from one form to other forms. If the power consumed, or *dissipated*, by a light bulb increases, it converts more electrical energy to light energy within the same period, and hence is brighter.

Cross-link

You learned power in Book 1 Chapter 3.

Power is denoted by P and measured in **watts** (**W**). Power of 1 W means 1 J of energy is converted every second.

1 W = 1 J s^{-1}

$$P = \frac{E}{t}$$

(P: power (W); E: energy converted (J); t: time (s))

Suppose a total charge Q passes through the bulb in time t (Fig 2.5b).

By $I = \frac{Q}{t}$ and $V = \frac{E}{Q}$,

$$E = VQ = V(It) = VIt$$

$$P = \frac{E}{t} = \frac{VIt}{t} = VI$$

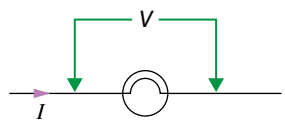


Fig 2.5b *I and V of a bulb.*

This is the power consumed but not the power of emitting visible light. This topic will not be included in this chapter.

The power delivered by an **electric source** or consumed by a **load** is given by

$$P = VI$$

(P: power (W); V: voltage (V); I: current (A))

electrical power 電功率　　dissipate 消耗　　watt (W) 瓦特

→ The simulation 'Series circuit' shows a series circuit consisting of two bulbs. Ss can change the value of resistance of each bulb, and observe how this affects the brightness of bulbs and the flow of current in the circuit.

→ The simulation 'Parallel circuit' shows a parallel circuit consisting of two bulbs. Ss can change the value of resistance of each bulb, and observe how this affects the brightness of bulbs and the flow of current in the circuit.

→ The simulation 'Complicated circuit' shows a circuit consisting of three bulbs. Ss can change the value of resistance of each bulb and observe how this affects the current, voltage and brightness of the bulbs.

The simulations are available on OUP web in PC version (.exe) only.

Simulation

For a **load** of resistance R, the equation $V = IR$ can be applied.

$$P = VI = (IR)I = I^2R \quad \text{or} \quad P = VI = V \times \left(\frac{V}{R}\right) = \frac{V^2}{R}$$

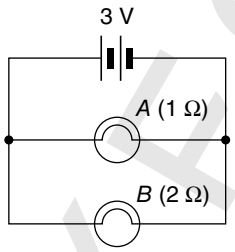
$$\boldsymbol{P = VI}$$
$$\boldsymbol{P = I^2R}$$
$$\boldsymbol{P = \frac{V^2}{R}}$$

Example 12 Brightness of two light bulbs

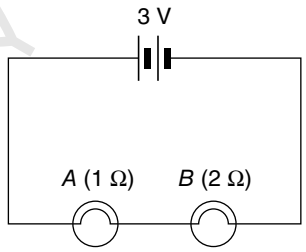
Light bulbs A and B have resistances of 1 Ω and 2 Ω respectively.

(a) Which bulb is brighter in the following cases?

(i)



(ii)



(b) Find the power delivered by the battery in the two cases.

Solution

(a) (i) $P_A = \frac{V^2}{R_A} = \frac{3^2}{1} = 9$ W

$P_B = \frac{V^2}{R_B} = \frac{3^2}{2} = 4.5$ W

$\therefore A$ is brighter.

(ii) $I = \frac{V}{R} = \frac{3}{1+2} = 1$ A

$P_A = I^2R_A = 1^2 \times 1 = 1$ W

$P_B = I^2R_B = 1^2 \times 2 = 2$ W

$\therefore B$ is brighter.

(b) (i) Current through battery $= I_A + I_B = \frac{V}{R_A} + \frac{V}{R_B} = \frac{3}{1} + \frac{3}{2} = 4.5$ A

$P = VI = 3 \times 4.5 = 13.5$ W

(ii) $P = VI = 3 \times 1 = 3$ W

In this book, we assume all light bulbs have the same efficiency.

Caution

A and B have the same V but different I.

Caution

A and B have the same I but different V.

Alternatively, by conservation of energy, $P = P_A + P_B$.

(i) $P = 9 + 4.5 = 13.5$ W

(ii) $P = 1 + 2 = 3$ W

Think about...

How would the power of A change if one more bulb is added in parallel in (a)(i) and in series in (a)(ii)?

We can see from the example that the power consumed by a load depends on how it is connected. Then what does the value '60 W' on the package of a light bulb mean (Fig 2.5c)?



Fig 2.5c *Package of a light bulb.*

This value is called the **rated value**. The rated power is the power consumed by the bulb when the bulb works at the rated voltage (see the green box). If the bulb works at another voltage, its actual power consumed will be different from the rated power.

By $R = \frac{V^2}{P}$, the 40-W bulb has a higher resistance.

For the two bulbs in Figure 2.5a (p.94), if they are connected in parallel, the 60-W bulb will be brighter. If they are connected in series, the 40-W bulb will be brighter.

Google Forms Microsoft Forms

Quick check 11

E-version in OUP Exercise Platform is also available.

1 Complete the following table about a light bulb connected to a battery.

	V / V	I / A	R / Ω	P / W
(a)	1.5	0.1		
(b)		0.2	15	
(c)	3			0.9

2 *True or false:*

(a) When two resistors are connected in series, the one with a higher resistance dissipates a higher power. (T/F)

(b) When two resistors are connected in parallel, the one with a higher resistance dissipates a higher power. (T/F)

(c) The power delivered by a battery changes when the total resistance of the circuit changes. (T/F)

(d) When more resistors are connected in series to a battery, the power delivered by the battery increases. (T/F)

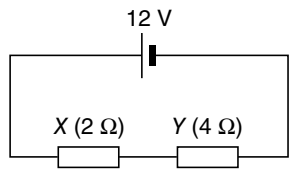
(e) When more resistors are connected in parallel to a battery, the power delivered by the battery increases. (T/F)

3 In each of the following cases, find

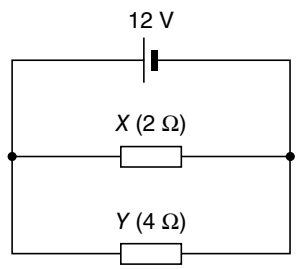
(i) the power dissipated by X and Y,

(ii) the power delivered by the battery.

(a)



(b)



rated value 額定值

Learn by practice 8

1 A light bulb consumes a power of 0.45 W when connected to a 1.5-V battery. Find its resistance.

2 Two 0.5-Ω resistors are connected in series to a battery. The power dissipated by each resistor is 2 W.
(a) Find the power delivered by the battery.
(b) Find the current drawn from the battery.
(c) Find the e.m.f. of the battery.

3 A 3-V battery delivers a power of 2.4 W. Find the current drawn from the battery.

4 A 2-Ω resistor and a 6-Ω resistor are connected in parallel to a battery. The current through the 2-Ω resistor is 1.5 A.
(a) Find the power dissipated by
(i) the 2-Ω resistor,
(ii) the 6-Ω resistor.
(b) Find the voltage across the 2-Ω resistor.

DSE goal
Examine the heating effect when current passes through conductor.

2 Heating effect of current

When a light bulb is connected to a battery, it gives out light and becomes hot. In fact, a conductor will always be heated up when current passes through it. This heating effect of current is used in electrical appliances for heating, such as space heaters (Fig 2.5d) and electric kettles. A heater with a higher power can heat up objects faster.

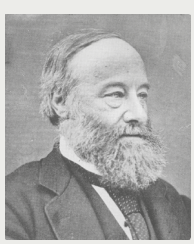


Fig 2.5d A space heater.

Note that if the power dissipated by a load is too high, it will become too hot and **burn out**. The circuit will then become **open**.

Know more

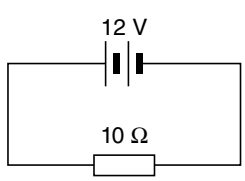
James Joule (1818–1889)



In 1840, *James Joule* suggested that heat could be generated by electric current. He measured that the heat produced in a fixed time period was proportional to I^2 and R. The unit of energy, the joule, is named after him.

Example 13 The power consumed by a heater

Janet designs a heater by connecting a 10-Ω resistor to a 12-V battery.



(a) Find the power consumed by the heater.

(b) Ken suggests that by $P = I^2R$, using a resistor of higher resistance can increase the power consumed by the heater. Comment on his suggestion.

Solution

(a) $P = \dfrac{V^2}{R} = \dfrac{12^2}{10} = 14.4$ W

(b) His suggestion is incorrect.

By $P = \dfrac{V^2}{R^2}$, since V is constant, $R \uparrow \Rightarrow P \downarrow$

Tip
By $I = \dfrac{V}{R}$, $R \uparrow \Rightarrow I \downarrow$
It is not appropriate to use $P = I^2R$ in this case since both I and R change.

James Joule 焦耳

Google Forms Microsoft Forms

Quick check 12

E-version in OUP Exercise Platform is also available.

1 Which of the following is a unit of power?

A J
B W
C W s
D kW h

2 *True or false:*

(a) When a current passes through a conductor, some electrical energy is always converted to thermal energy. (T/F)

(b) The power consumed by a heater will increase if its resistance decreases. (T/F)

Exercise 2.5

★**1** Which of the following statements is **incorrect**?

A For two identical light bulbs, the one with a larger current passing through dissipates a higher power.

B For two identical light bulbs, the one with a larger voltage across it dissipates a higher power.

C For two light bulbs with the same current passing through them, the one with a higher resistance dissipates a higher power.

D For two light bulbs with the same voltage across them, the one with a higher resistance dissipates a higher power.

★**2** The switch in the circuit is open initially (Fig a).

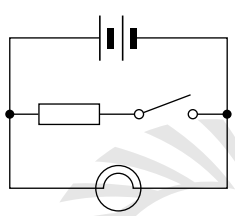


Fig a

What will happen to the bulb when the switch is closed?

A It becomes brighter.
B It becomes dimmer.
C It remains unchanged.
D Cannot be determined

★**3** A student designs a heater by connecting a resistance wire X to an electric source. Explain how the power consumed by the heater will change if another resistance wire is connected

(a) in series to X,

(b) in parallel to X.

★**4** Figure b shows 3 resistors connected to a battery. The total power dissipated by the 3 resistors is 12 W.

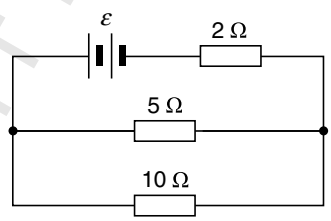


Fig b

(a) Find the equivalent resistance of the circuit.

(b) Find the e.m.f. ε.

(c) Calculate the power dissipated by each resistor.

★**5** In Figure c, circuits 1, 2 and 3 consist of identical light bulbs of resistance 1 Ω. A 6-V battery is connected to each of the circuits.

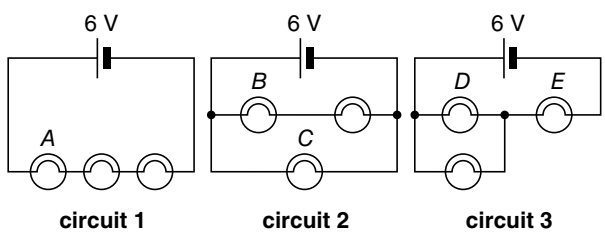


Fig c

(a) Calculate the current through the battery in each circuit.

(b) Calculate the power dissipated by bulbs A–E.

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Key concepts 2

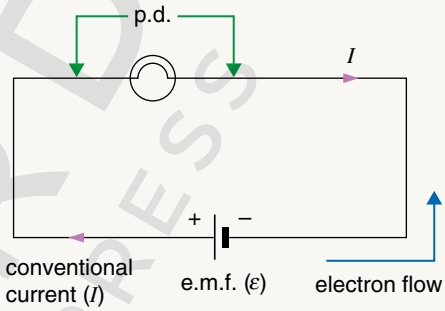
1 Electric circuit

- Charges flow in a closed circuit with an electric source.
- Current I

current (A) $\rightarrow I = \frac{Q}{t}$ ← charge (C), time (s)

- Voltage (e.m.f. or p.d.) V

voltage (V) $\rightarrow V = \frac{E}{Q}$ ← energy converted (J), charge (C)

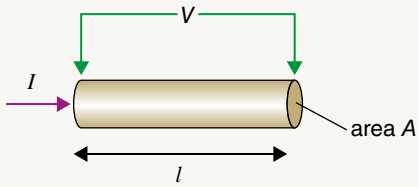


2 Resistance

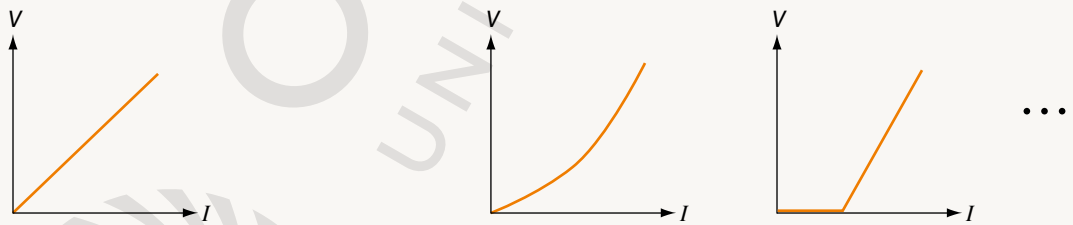
- Resistance is a measure of an object's opposition to the flow of current.

resistance (Ω) $\rightarrow R = \frac{V}{I}$

resistivity of material of wire (Ω m), length of wire (m), cross-sectional area of wire (m²) $\rightarrow R = \rho \frac{l}{A}$



3 Ohmic and non-ohmic conductors



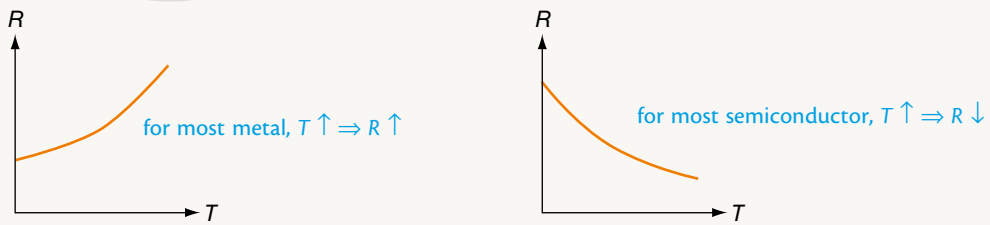
ohmic conductor:
- obey Ohm's law ($V \propto I$)
- V–I graph is a straight line passing through the origin
- R ($= \frac{V}{I}$) constant

} at constant temperature

non-ohmic conductor:
- not obey Ohm's law
- V–I graph not a straight line passing through the origin
- R ($= \frac{V}{I}$) not constant

} at constant temperature

4 Effect of temperature on resistance



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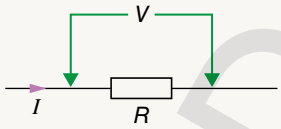
5 Power

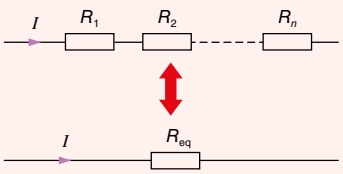
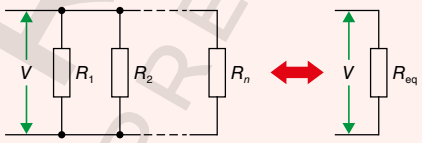
- Power is the rate at which energy is converted from one form to other forms.

power (W) ⋯⋯▸ $P = VI$

$P = I^2R$

$P = \dfrac{V^2}{R}$



Resistors in series	**Resistors in parallel**
	
$R_{eq} = R_1 + R_2 + \ldots + R_n$ R_{eq} larger than any single resistor	$\dfrac{1}{R_{eq}} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \ldots + \dfrac{1}{R_n}$ R_{eq} smaller than any single resistor
$V_1 = IR_1$, $V_2 = IR_2$, … larger $R \Rightarrow$ larger V	$V_1 = V_2 = \ldots = V$
$I_1 = I_2 = \ldots = I$	$I_1 = \dfrac{V}{R_1}$, $I_2 = \dfrac{V}{R_2}$, … larger $R \Rightarrow$ smaller I
$P_1 = I^2R_1$, $P_2 = I^2R_2$, … larger $R \Rightarrow$ larger P total $P = P_1 + P_2 + \ldots + P_n$	$P_1 = \dfrac{V^2}{R_1}$, $P_2 = \dfrac{V^2}{R_2}$, … larger $R \Rightarrow$ smaller P total $P = P_1 + P_2 + \ldots + P_n$

Key terms

1 ammeter 安培計 p.60
2 ampere (A) 安培 p.56
3 electric current 電流 p.56
4 electric potential 電勢 p.66
5 electrical power 電功率 p.94
6 electromotive force (e.m.f.) 電動勢 p.65
7 equivalent resistance 等效電阻 p.85
8 in parallel 並聯 p.60
9 in series 串聯 p.60
10 ohm (Ω) 歐姆 p.73
11 Ohm's law 歐姆定律 p.76
12 potential difference (p.d.) 電勢差 p.65
13 resistance 電阻 p.73
14 resistivity 電阻率 p.80
15 volt (V) 伏特 p.65
16 voltage 電壓 p.66
17 voltmeter 伏特計 p.68
18 watt (W) 瓦特 p.94

Revision 2

Multiple-choice questions

★1 2.1 A cow stands in front of an electric fence (Fig a).



Fig a

It receives a current of 20 mA when it touches the fence. How many electrons pass through it in 0.5 s? The charge of an electron is -1.6×10^{-19} C.

A 2.5×10^{16} B 6.25×10^{16}

C 6.25×10^{17} D 2.5×10^{17}

★2 2.1 Lightning (Fig b) can be considered as a flow of current. A typical lightning flash lasts for 0.1 s and transfers 10^{11} J of energy. If the typical p.d. between the thundercloud and the ground is 10^8 V, estimate the size of the current in a typical lightning flash.



Fig b

A 1 A B 100 A

C 10 000 A D 10^{10} A

★3 2.3 A p.d. is applied to a copper wire. Which of the following changes can increase the amount of charge passing through the wire every second?

(1) Increase the applied p.d.

(2) Increase the length of the wire.

(3) Reduce the diameter of the wire.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

★4 2.4 Two resistors R_1 and R_2 are connected in series to a battery (Fig c).

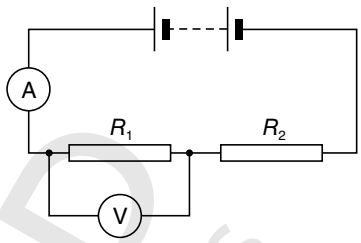


Fig c

The resistance of R_2 is 20 Ω. The readings of the voltmeter and the ammeter are 1 V and 0.25 A respectively. What is the e.m.f. of the battery?

A 2 V B 4 V

C 5 V D 6 V

★5 2.4 In the circuit (Fig d), the switch is initially open.

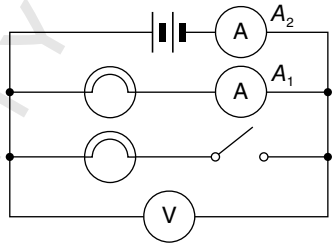


Fig d

How will the readings of the ammeters and the voltmeter change once the switch is closed?

(1) The reading of A_1 will be unchanged.

(2) The reading of A_2 will decrease.

(3) The voltmeter reading will be unchanged.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

★6 2.4 A fixed resistor r and a variable resistor R are connected as shown (Fig e).

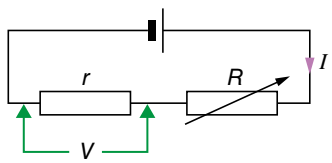


Fig e

How do V and I change when the resistance of R decreases?

	V	I
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

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★7 2.3 An eureka wire is connected to a battery. The p.d. V across the eureka wire and the current I passing through it are measured. A graph of I against V is obtained (Fig f).

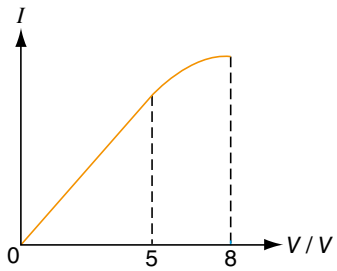


Fig f

How does the resistance of the eureka wire change when V is increased from 0 to 5 V and from 5 V to 8 V?

	From 0 to 5 V	From 5 V to 8 V
A	increases	increases
B	remains unchanged	increases
C	remains unchanged	decreases
D	decreases	decreases

★8 2.4 Three resistors X, Y and Z have resistance 1 Ω, 2 Ω and 3 Ω respectively. They are connected in series to a battery. The current through X, Y and Z are I_X, I_Y and I_Z respectively. The p.d. across X, Y and Z are V_X, V_Y and V_Z respectively. Find $I_X : I_Y : I_Z$ and $V_X : V_Y : V_Z$.

	$I_X : I_Y : I_Z$	$V_X : V_Y : V_Z$
A	1 : 1 : 1	1 : 2 : 3
B	1 : 1 : 1	6 : 3 : 2
C	1 : 2 : 3	1 : 1 : 1
D	6 : 3 : 2	1 : 1 : 1

★9 2.4 Three resistors X, Y and Z have resistance 2 Ω, 3 Ω and 4 Ω respectively. They are connected in parallel to a battery. The current through X, Y and Z are I_X, I_Y and I_Z respectively. The p.d. across X, Y and Z are V_X, V_Y and V_Z respectively. Find $I_X : I_Y : I_Z$ and $V_X : V_Y : V_Z$.

	$I_X : I_Y : I_Z$	$V_X : V_Y : V_Z$
A	1 : 1 : 1	2 : 3 : 4
B	1 : 1 : 1	6 : 4 : 3
C	2 : 3 : 4	1 : 1 : 1
D	6 : 4 : 3	1 : 1 : 1

★10 Misc Figure g shows the V–I graphs of two resistors X and Y.

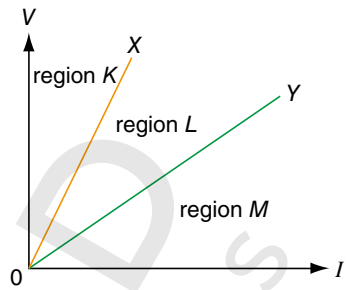
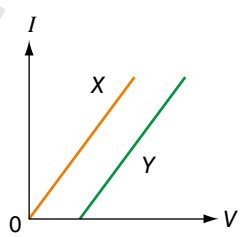


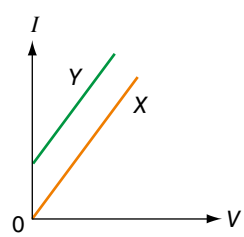
Fig g

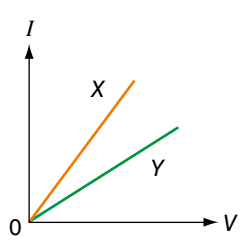
The two resistors are then connected together to form a resistor Z. In which region will the V–I graph of Z be if the two resistors are connected in series and in parallel respectively to form Z?

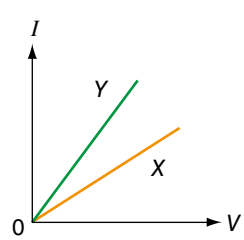
	In series	In parallel
A	K	L
B	K	M
C	M	L
D	M	K

★11 2.3 Iron wires X and Y have the same thickness but X is longer than Y. Which of the following figures correctly shows their I–V graphs?

A


B


C


D


★12 2.4 James is given a certain number of identical bulbs and a 6-V battery. When he connects all the bulbs in series to the battery, the current drawn from the battery is 0.15 A. When he connects all the bulbs in parallel, the current drawn from the battery is 2.4 A. Find the number of bulbs.

A 4 B 6

C 16 D 256

★13 2.3 The V–I graph of a conductor X is as shown (Fig h).

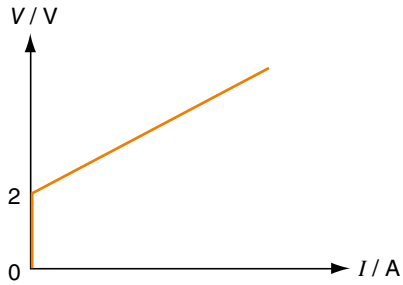


Fig h

Which of the following statements is/are correct?

(1) When V is larger than 2 V, the resistance of X decreases when V increases.

(2) When V is larger than 2 V, $V \propto I$.

(3) When V = 1 V, the resistance of X = 0.

A (1) only

B (3) only

C (1) and (2) only

D (2) and (3) only

★★14 2.5 A power supply with constant e.m.f. is connected to a fixed resistor M and a variable resistor N in series (Fig i).

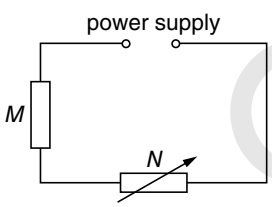
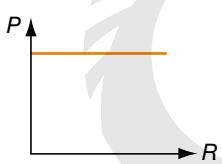
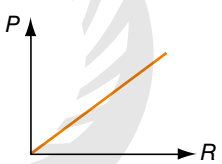
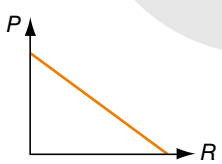


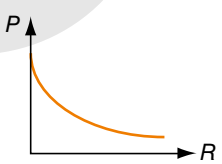
Fig i

Which of the following graphs correctly show how the power P dissipated by M changes with the resistance R of N?

A 

B 

C 

D 

▶ Refer p.86, 95

★★15 Misc A resistance wire consists of two segments. The lengths of the segments are l_1 and l_2, and their resistivities are ρ_1 and ρ_2 where $\rho_1 > \rho_2$. The wire is connected in a circuit as shown in Figure j. Suppose the thickness of the wire is uniform.

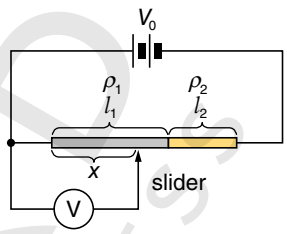
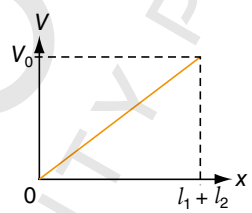
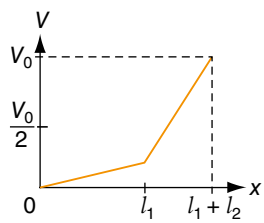
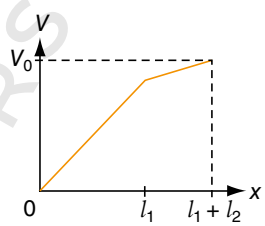


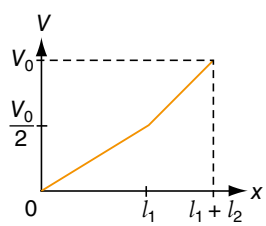
Fig j

Which of the following graphs best represents how the voltmeter's reading V varies with the distance x?

A 

B 

C 

D 

▶ Refer p.79, 80

★★16 Misc A resistor becomes hot when a voltage is applied across it. Thermal equilibrium is reached when the power dissipated by it is equal to its rate of heat loss to the surroundings. When a voltage of 1 V is applied, thermal equilibrium is reached at 50 °C. What is the voltage applied if thermal equilibrium is reached at 125 °C? Assume that the room temperature remains at 25 °C. Also, the resistance of the heating element is constant and its rate of heat loss is directly proportional to the temperature difference between it and the environment.

A 2 V

B 4 V

C 8 V

D 16 V

▶ Refer p.95

Q17 Exam Report: It seemed that some candidate did not fully understand the definition of 'potential difference' as more than one-third of them wrongly held that statement (3) is incorrect.

Misc **17** **HKDSE 2016 Paper 1A Q26**

Two filament light bulbs X and Y are connected in parallel to a dry cell. X is brighter than Y. Which statements are correct?

(1) In 1 s, the number of charges flowing through X is greater than that flowing through Y.

(2) In 1 s, the electrical energy dissipated by X is greater than that dissipated by Y.

(3) For every unit charge passing, the electrical energy dissipated by X is equal to that dissipated by Y.

A (1) and (2) only (37%)

B (1) and (3) only (17%)

C (2) and (3) only (13%)

D (1), (2) and (3) (33%)

2.3 **18** **HKDSE 2018 Paper 1A Q23**

Figure k shows the current-voltage (I–V) relationship of a conductor.

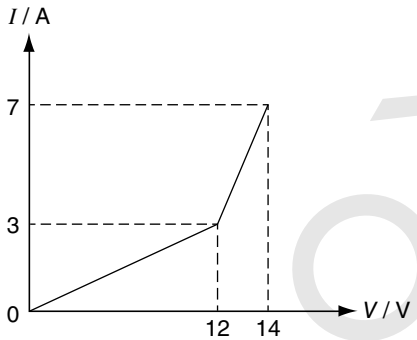


Fig k

Which statement is **incorrect**?

A When the voltage across the conductor is less than 12 V, it obeys Ohm's law.

B When the voltage across the conductor exceeds 12 V, its resistance begins to decrease.

C The resistance of the conductor is 0.5 Ω when the current flowing through it is 5 A. (69%)

D The resistance of the conductor is 2 Ω when the voltage across it is 14 V.

2.3 **19** **HKDSE 2018 Paper 1A Q25**

Two wires X and Y of the same length are made from different materials. The radius of X is half that of Y. When the two wires are connected in parallel to the same power supply, the current flowing through each wire is the same. What is the ratio of the resistivity of the material of X to that of Y?

A 1 : 4 (51%)

B 4 : 1

C 1 : 2

D 2 : 1

2.3 **20** **HKDSE 2021 Paper 1A Q24**

Figure l shows a metallic cube of side length x. How is its resistance R between any two opposite faces related to x?

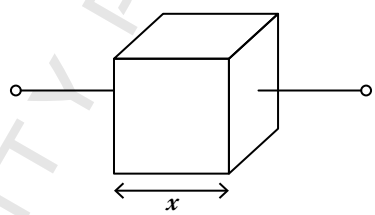


Fig l

A $R \propto \frac{1}{x}$ (45%)

B $R \propto x$ (15%)

C $R \propto x^2$ (15%)

D $R \propto \frac{1}{x^2}$ (25%)

Exam Report: Candidates choosing option D may only have considered the cross-section of the cube and overlooked the contribution of its length.

Conventional questions

★ **21** 2.2 A student connects the circuit below (Fig m). Bulbs X and Y are connected in parallel. He wants to measure the e.m.f. of the battery box and the current through bulb Y.

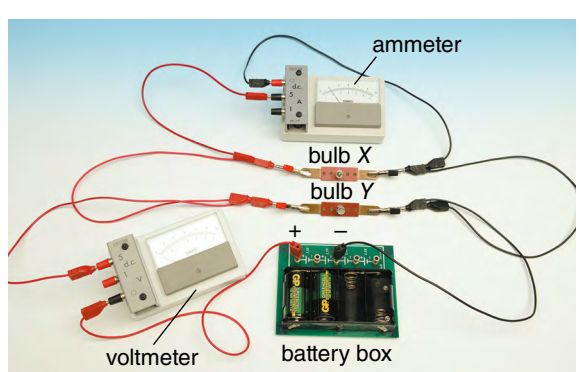


Fig m

(a) State three mistakes in the circuit. (3 marks)

(b) Draw a circuit diagram to show the correct circuit for the measurements. (3 marks)

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★**22** 2.4 A variable resistor R_1 and a fixed resistor R_2 are connected in series to a battery of e.m.f. ε (Fig n).

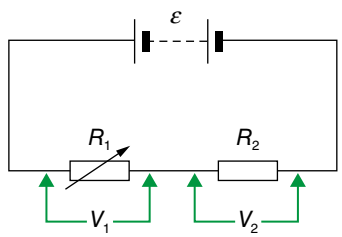


Fig n

(a) If R_1 is larger than R_2, compare V_1 and V_2. (1 mark)

(b) Can we make V_2 larger by changing the resistance of R_1? If yes, how? (2 marks)

(c) Given that ε = 12 V and R_2 = 6 Ω, find V_1 when

(i) R_1 = 2 Ω, (2 marks)

(ii) R_1 = 12 Ω. (1 mark)

★**23** 2.4 Three light bulbs and five ammeters are connected to a battery (Fig o). The reading of ammeters A_1, A_2 and A_4 are 4 A, 1 A and 1 A respectively.

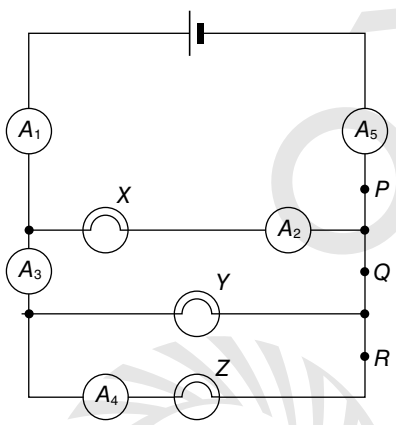


Fig o

(a) Draw arrows to show the current direction at points P, Q and R. (2 marks)

(b) Indicate the + terminal of each ammeter. (2 marks)

(c) Which ammeter(s) can measure the current through bulb X? (1 mark)

(d) Find the reading of ammeters A_3 and A_5. (2 marks)

(e) If the resistance of bulb X is 10 Ω, find the resistance of bulbs Y and Z. (3 marks)

★**24** 2.4 Three resistors are connected to a battery as shown (Fig p).

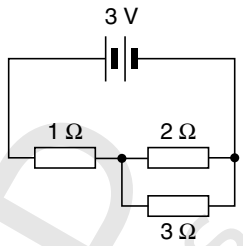


Fig p

(a) Find the equivalent resistance of the circuit. (2 marks)

(b) Find the current through the 1-Ω resistor. (2 marks)

(c) Find the p.d. across each resistor. (3 marks)

(d) Find the current through the 2-Ω and 3-Ω resistors. (3 marks)

★**25** 2.3 (a) Describe, with the aid of a circuit diagram, a method to find how the resistance R of a metal wire of uniform thickness varies with its length l. (3 marks)

(b) A graph of R against l for two metal wires of the same material but different thicknesses are plotted in Figure q.

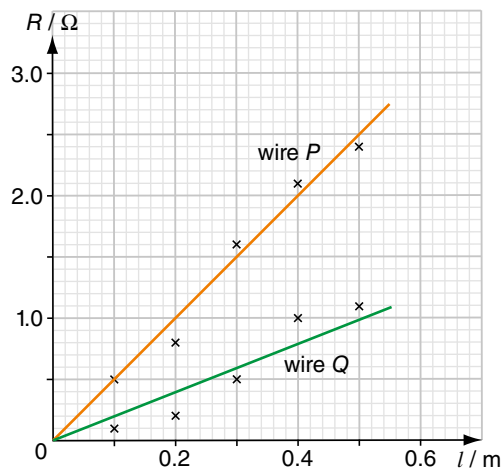


Fig q

(i) State the relationship between R and l according to the graphs. (1 mark)

(ii) For the same length of wire P and wire Q, which wire has a higher resistance? (1 mark)

(iii) Which wire is thicker? Explain your answer. (3 marks)

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★**26** 2.5 Jimmy has three identical small bulbs *W*, *X* and *Y*, and a light bulb *Z* which has a resistance different from the others. He connects these bulbs to make a string of lights (Fig r). *Z* appears dimmer than the others. Take the resistances of the bulbs to be constant.

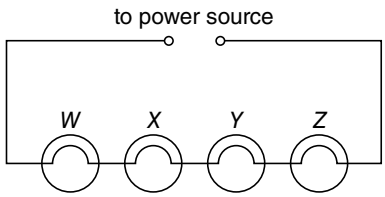


Fig r

(a) What can you say about the resistance of *Z*? Explain your answer. (2 marks)

(b) If the bulbs are connected in parallel,

(i) would *W* become brighter or dimmer? (1 mark)

(ii) would *Z* be dimmer or brighter than the others? (1 mark)

(c) Jimmy claims that he can increase the power consumed by the string of lights by adding more small bulbs in series to the original bulbs. Comment on his claim. (3 marks)

★**27** 2.1 A light conducting ball suspended by an insulating thread is placed between two vertical metal plates connected to a power supply (Fig s). At first, the ball is attracted by the negative plate. It is repelled by the plate after touching. It then moves towards the positive plate and is repelled again after touching. The ball moves between the plates and touches each plate 40 times in a minute. The current registered by the galvanometer is 0.096 μA.

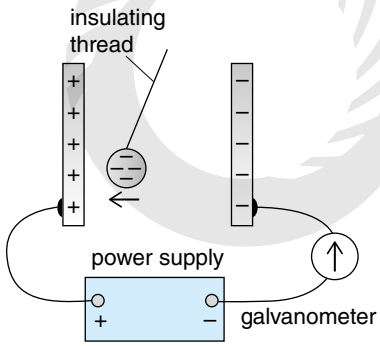


Fig s

(a) Explain why the ball moves between the plates. (3 marks)

(b) Draw the direction of the current in Figure s. (1 mark)

(c) Calculate the charge carried by the ball each time the ball travels from the negative plate to the positive one. (2 marks)

★**28** Misc A student designs a thermometer by connecting a semiconductor *X* to a 100-Ω resistor, an ammeter and a 3-V battery. *X* and the 100-Ω resistor can be connected in series (Fig t) or in parallel (Fig u).

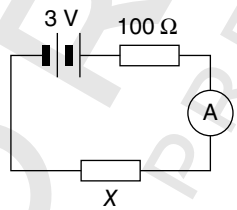


Fig t

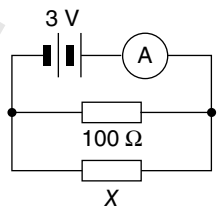


Fig u

The resistance *R* of *X* changes with temperature *T* as shown in Figure v.

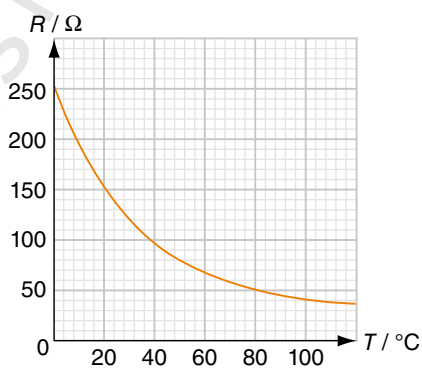


Fig v

(a) Find the current through *X* at $T = 80$ °C in the two circuits. (3 marks)

(b) Find the p.d. across *X* at $T = 80$ °C in the two circuits. (3 marks)

(c) In the two circuits, how does the ammeter reading change when *T* increases? (2 marks)

★**29** Misc Three light bulbs of resistance 1 Ω, 2 Ω and 3 Ω are connected to a 6-V battery. Each bulb can be turned on and off independently and the brightness is not affected by the other two bulbs.

(a) Draw a circuit diagram to show how the bulbs are connected. (2 marks)

(b) Find the power dissipated by each bulb when it is turned on. (4 marks)

★30 2.3 A student carries out an experiment to investigate whether a bulb filament obeys Ohm's law. He connects the apparatus as shown (Fig w). The current sensor and voltage sensor are used to measure the current through the bulb and the voltage across the bulb.

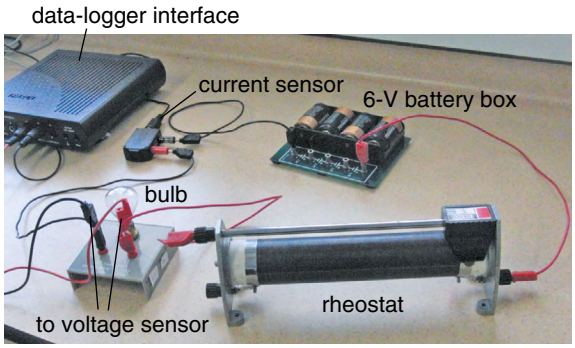


Fig w

He lights up the bulb for some time so that the bulb filament reaches a steady temperature. He then starts data-logging and slides the moving contact of the rheostat quickly from one end to the other. After that, he changes the connection of the battery box from 6 V to 4.5 V and repeats the measurement. Figure x shows the results obtained.

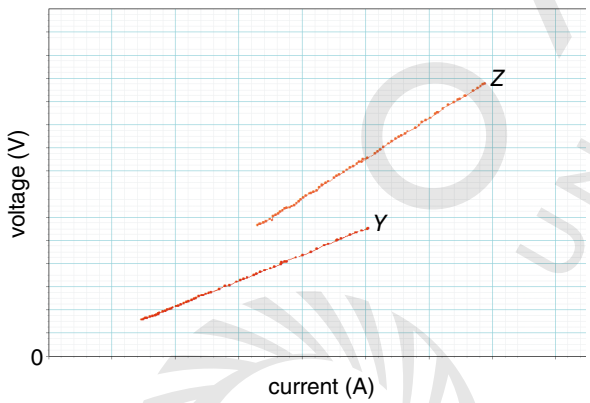


Fig x

(a) Why does the sliding contact have to be moved quickly? (2 marks)

(b) Does the filament obey Ohm's law? State the reason for your answer. (2 marks)

(c) Determine which graph in Figure x corresponds to using the 4.5-V battery box. Explain briefly. (2 marks)

★★31 Misc Figure y shows the voltage–current (V–I) characteristics of two electrical components X and Y at constant temperature.

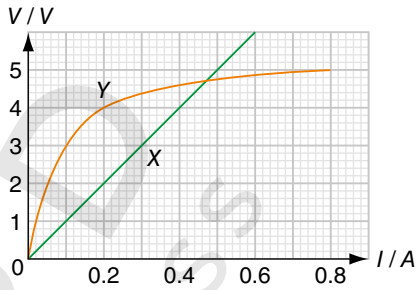


Fig y

(a) (i) Find the resistance of X. (1 mark)

(ii) If X is an alloy wire of area 4.9×10^{-8} m^2 and resistivity 1.1×10^{-6} Ω m, find its length. (2 marks)

(b) Explain, with reference to Figure y, whether X and Y are ohmic or not. (2 marks)

(c) X and Y are connected in parallel as shown (Fig z). The e.m.f. of the cell is 3 V.

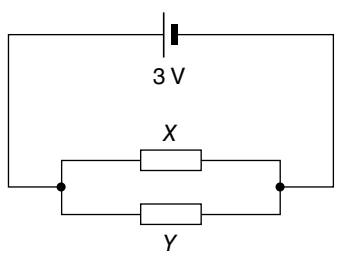


Fig z

(i) Find the equivalent resistance of X and Y. (2 marks)

(ii) Find the total power dissipated by the circuit. (1 mark)

(d) X and Y are now connected in series as shown (Fig aa). The e.m.f. of the cell is 6 V.

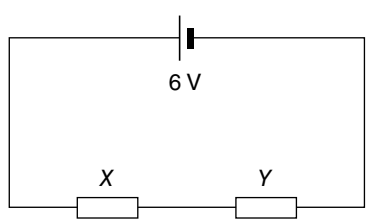


Fig aa

(i) Find the current drawn from the battery. [Hint: $V_X + V_Y = 6$ V] (2 marks)

(ii) Find the power dissipated by X and Y. (3 marks)

Refer p.77

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2.3 32 HKDSE 2017 Paper 1B Q8

A student uses the following apparatus to measure the resistance of a tungsten filament light bulb. A battery, a switch, a variable resistor, an ammeter, a voltmeter, a light bulb

(a) Figure ab shows an incomplete circuit for the experiment. The '+' symbol represents the positive terminal of the ammeter. Use suitable circuit symbols to complete the circuit and mark the positive terminal of the voltmeter with '+'. (3 marks)

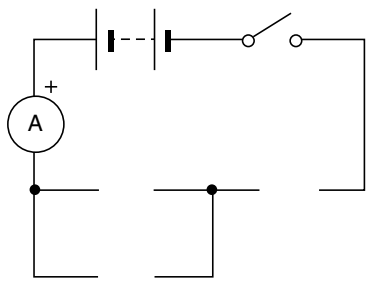


Fig ab

Table a below and Figure ac show the results obtained.

Voltage across the light bulb V / V	Current I / mA
0	0
0.1	76
0.2	112
0.3	126
0.4	133
0.5	139
1.0	170
2.0	226
3.0	273

Table a

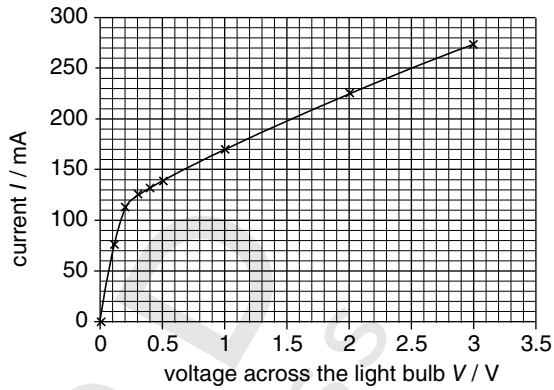


Fig ac

(b) Briefly explain the variation of the resistance of the light bulb with the voltage across the light bulb. (2 marks)

(c) The student claims that since the resistance of light bulb is not a constant, the equation $R = V/I$ cannot be used to calculate the resistance of the light bulb. Briefly explain why his claim is wrong. (1 mark)

(d) Determine the resistance of the light bulb at $V = 0.1$ V and 2.5 V. (3 marks)

(e) It is given that the cross-sectional area of the tungsten filament in the light bulb is 1.66×10^{-9} m^2, and the resistivity of tungsten at room temperature is about 5.6×10^{-8} Ω m. Estimate the length of the tungsten filament in the light bulb using the appropriate resistance found in (d). (3 marks)

Exam Report: This question tested candidates' understanding of the relationship among voltage, current and resistance in a circuit. Candidates' performance was poor. Part (a) was well answered. In (b), few were able to state explicitly that the temperature increase led to the increase in resistance. Not many pointed out that by definition $R = V/I$ in part (c). Although most were able to obtain the correct resistances in (d), not many realized that the one corresponding to room temperature should be employed to find the length of the tungsten filament in part (e).

Know more

Electric potential (p.66)

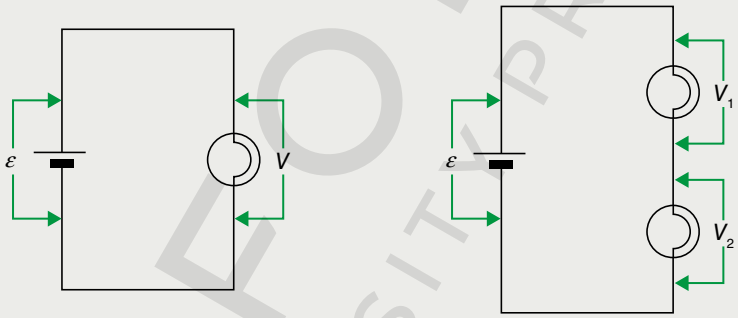
The p.d. across two points is the difference in the electric potentials at these points. This is like calculating the change in gravitational potential energy (GPE) when an object moves from one point to another. In the calculation, the reference level that GPE = 0 J can be chosen arbitrarily (see Book 2 Chapter 6.2). Similarly, we can choose any position in a circuit to be the reference point that the electric potential = 0 V. The **p.d. across any two points is independent of the reference point.**

In practice, if a point in a circuit is earthed, we will assign the potential of this point to be 0 V.

Voltage and conservation of energy (p.69)

Consider the electric potential energy of the charges.

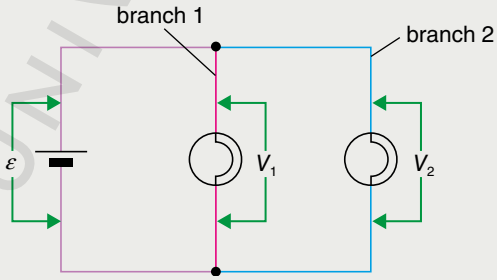
For series connection:



Energy gained from battery = $Q\varepsilon$
Energy lost at load = QV
$\varepsilon = V$

Energy gained from battery = $Q\varepsilon$
Energy lost at load = $QV_1 + QV_2$
$\varepsilon = V_1 + V_2$

For parallel connection:



After gaining energy from the battery, a charge Q can only flow through branch 1 or 2.

If Q flows through branch 1,

energy gained = energy lost

$$Q\varepsilon = QV_1$$

$$\varepsilon = V_1$$

Similarly, if Q flows through branch 2,

$$\varepsilon = V_2$$

$$\therefore \varepsilon = V_1 = V_2$$

Batteries contain corrosive materials. Do not disassemble a battery yourself.

Know more

Cells in series (p.70)

A 9-V battery consists of 6 cylindrical 1.5-V cells connected in series.



An electric eel has electricity-generating cells called electrocytes. Each electrocyte acts like a 0.15-V battery. Thousands of electrocytes are connected in series to produce a voltage as high as 600 V.



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